chapter 7

Clinical Manifestations and Useful Tests

Normally, the pleural space contains only a few milliliters of pleural fluid. If fluid in the pleural space is detected on a radiologic examination, it is abnormal. Many conditions can be associated with pleural fluid accumulation (Table 8.1). When pleural fluid is detected, an effort should be made to determine which of the many conditions listed in Table 8.1 is responsible. In this chapter, the clinical manifestations of pleural effusions are first discussed. Then, the various tests used in the differential diagnosis of pleural effusions are reviewed. In Chapter 8, recommendations are given for a systematic approach to the patient with an undiagnosed pleural effusion.

CLINICAL MANIFESTATIONS

The presence of moderate-to-large amounts of pleural fluid produces symptoms and characteristic changes on physical examination.

Symptoms

The symptoms of a patient with a pleural effusion are mainly dictated by the underlying process causing the effusion. Many patients have no symptoms referable to the effusion. When symptoms are related to the effusion, they arise either from inflammation of the pleura, from compromise of pulmonary mechanics, from interference with gas exchange, or, on rare occasions, from decreased cardiac output. A pleural effusion associated with pleuritic chest pain indicates inflammation of the pleura, specifically, the parietal pleura as the visceral pleura does not have pain fibers. Some patients with pleural effusions experience a dull, aching chest pain rather than pleuritic chest pain. This symptom is very suggestive that the patient has pleural malignancy (1). The presence of either pleuritic chest pain or dull, aching chest pain indicates that the parietal pleura is probably involved and that the patient has an exudative pleural effusion.

Ordinarily, the pain associated with pleural disease is well localized and coincides with the affected area of the pleura because the parietal pleura is innervated mostly by the intercostal nerves. At times, however, pleuritic pain is referred to the abdomen because intercostal nerves are also distributed to the abdomen. A notable exception to the localization of the pain occurs when the central portion of the diaphragmatic pleura is involved. The nerve supply to this portion of the parietal pleura is the phrenic nerve; therefore, inflammation of the central portion of the diaphragm is referred to the tip of the ipsilateral shoulder. Pleuritic pain felt simultaneously in the lower chest and ipsilateral shoulder is pathognomonic of diaphragmatic involvement.

A second symptom of pleural effusion is a dry, nonproductive cough. The mechanism producing the cough is not clear, although it may be related to pleural inflammation. Alternately, lung compression by the fluid may bring opposing bronchial walls into contact, stimulating the cough reflex.

The third symptom of pleural effusion is dyspnea. A pleural effusion acts as a space-occupying process in the thoracic cavity and therefore reduces all subdivisions of lung volumes. Small-to-moderate-sized pleural effusions displace rather than compress the lung and have little effect on pulmonary function (2). Larger pleural effusions obviously cause a significant reduction in lung volumes, but the improvement in pulmonary function following therapeutic thoracentesis is much less than what one would anticipate. We obtained spirometry before and 24 hours after thoracentesis in 26 patients from whom a mean of 1,740 mL pleural fluid was withdrawn (3). In these patients, the mean vital capacity improved 410 ± 390 mL. Patients in this study, with higher pleural pressures after the removal of 800 mL pleural fluid and patients with smaller decreases in the pleural pressure after the removal of 800 mL pleural fluid, had greater improvements in the forced vital capacity (FVC) after thoracentesis.

Associated parenchymal disease probably explains this small increase in pulmonary function following therapeutic thoracentesis. The degree of dyspnea is frequently out of proportion to the size of the pleural effusion. Often, this feature is the result of compromised diaphragmatic function due to the weight of fluid on the diaphragm. At times, the diaphragm becomes inverted and this usually results in disproportionate dyspnea (4). Either pleuritic chest pain, with the resultant splinting, or concomitant parenchymal disease can also be responsible for the disproportionate dyspnea. When the pleural effusion is large, ventricular filling may be impeded, leading to decreased cardiac output and dyspnea (5). Arterial blood gases usually remain at clinically acceptable levels whatever the size of the effusion (6) because of the reflex reduction in perfusion to the lung underlying the effusion.

Physical Examination

When a patient presents with chest symptoms, the physical examination is useful in suggesting whether pleural fluid is present (7). In such an examination, attention should be paid to the relative sizes of the hemithoraces and the intercostal spaces. If the pleural pressure is increased on the side of the effusion, that hemithorax will be larger, and the usual concavity of the intercostal spaces will be blunted or even convex. In contrast, if the pleural pressure on the side of the effusion is decreased, as with obstruction of a major bronchus or a trapped lung, the ipsilateral hemithorax will be smaller, and the normal concavity of the intercostal spaces will be exaggerated. In addition, with inspiratory efforts, the intercostal spaces retract. Enlargement of the hemithorax with bulging of the intercostal spaces is an indication for therapeutic thoracentesis to relieve the increased pleural pressure. Signs of decreased pleural pressure are a relative contraindication to therapeutic thoracentesis because the decreased pleural pressure can lead to reexpansion pulmonary edema (8). Of course, in many patients with pleural effusions, the hemithoraces are equal in size and the intercostal spaces are normal.

Palpation of the chest in patients with pleural effusions is useful in delineating the extent of the effusion. In areas of the chest where pleural fluid separates the lung from the chest wall, tactile fremitus is absent or attenuated because the fluid absorbs the vibrations emanating from the lung. Tactile fremitus is much more reliable than percussion for identifying both the upper border of the pleural fluid and the proper site to attempt a thoracentesis. With a thin rim of fluid, the percussion note may still be resonant, but the tactile fremitus is diminished. Palpation may also reveal that the cardiac point of maximum impulse is shifted to one side or the other. With large left pleural effusions, the cardiac point of maximum impulse may not be palpable. In patients with pleural effusions, the position of the trachea should always be ascertained because it indicates the relationship between the pleural pressures in the two hemithoraces.

The percussion note over a pleural effusion is dull or flat. The dullness is maximum at the lung bases where the thickness of the fluid is the greatest. As mentioned earlier, however, the percussion note may not be duller if only a thin rim of fluid is present. Light percussion is better than heavy percussion for identifying small amounts of pleural fluid. If the dullness to percussion shifts as the position of the patient is changed, one can be almost certain that free pleural fluid is present (9).

Auscultation over the pleural fluid characteristically reveals decreased or absent breath sounds. Near the superior border of the fluid, however, breath sounds may be accentuated and may take on a bronchial characteristic. This phenomenon has been attributed to increased conductance of breath sounds through the partially atelectatic lung compressed by the fluid (10). This accentuation of breath sounds does not mean that an associated parenchymal infiltrate is present. Auscultation may also reveal a pleural rub. Pleural rubs are characterized by coarse, creaking, leathery sounds most commonly heard during the latter part of inspiration and the early part of expiration, producing a to-and-fro pattern of sound. Pleural rubs, caused by the rubbing together of the roughened pleural surfaces during respiration, are often associated with local pain on breathing that subsides with breath-holding. Pleural rubs often appear as pleural effusions diminish in size, either spontaneously or as a result of treatment, because the pleural fluid is no longer present between the roughened pleural surfaces.

It is important to realize that an elevated hemidiaphragm can produce all the classic physical findings associated with a pleural effusion. Obviously, the chest is not the only structure that should be examined when evaluating a patient with a pleural effusion; clues to the origin of the effusion are often present elsewhere. The effusion is probably due to congestive heart failure (CHF) if the patient has cardiomegaly, neck vein distension, or peripheral edema. Signs of joint disease or subcutaneous nodules suggest that the pleural effusion is due to rheumatoid disease or lupus erythematosus. An enlarged, nontender nodular liver or the presence of hypertrophic osteoarthropathy suggests metastatic disease, as do breast masses or the absence of a breast. Abdominal tenderness suggests a subdiaphragmatic process, whereas tense ascites suggests cirrhosis and a hepatothorax. Lymphadenopathy suggests lymphoma, metastatic disease, or sarcoidosis.

SEPARATION OF TRANSUDATIVE FROM EXUDATIVE EFFUSIONS

The accumulation of clinically detectable quantities of pleural fluid is distinctly abnormal. A diagnostic thoracentesis (see Chapter 28) should be attempted whenever the thickness of pleural fluid on ultrasound or the decubitus radiograph is greater than 10 mm or whenever loculated pleural fluid is demonstrated with ultrasound unless the etiology of the effusion is known. A properly performed diagnostic thoracentesis takes less than 10 minutes and should cause no more morbidity than a venipuncture. The information available from examination of the pleural fluid is invaluable in the management of the patient.

Pleural effusions have classically been divided into transudates and exudates (11). A transudative pleural effusion develops when the systemic factors influencing the formation or absorption of pleural fluid are altered so that pleural fluid accumulates. The pleural fluid is a transudate. The fluid may originate in the lung, the pleura, or the peritoneal cavity (12). The permeability of the capillaries to proteins is normal in the area where the fluid is formed. Examples of conditions producing transudative pleural effusions are left ventricular failure producing increased pulmonary interstitial fluid and a resulting pleural effusion, ascites due to cirrhosis with movement of fluid through the diaphragm, and decreased serum oncotic pressure with hypoproteinemia.

In contrast, an exudative pleural effusion develops when the pleural surfaces or the capillaries in the location where the fluid originates are altered such that fluid accumulates. The pleural fluid is an exudate. The most common causes of exudative pleural effusions are pleural malignancy, parapneumonic effusions, and pulmonary embolism.

The first question to ask in assessing a patient with a pleural effusion is whether that effusion is a transudate or an exudate. If the effusion is a transudate, no further diagnostic pleural procedures are necessary, and therapy is directed to the underlying CHF, cirrhosis, or nephrosis. Alternately, if the effusion proves to be an exudate, a more extensive diagnostic investigation is indicated to delineate the cause of the effusion. It has been shown that pulmonary specialists are not very accurate at doing this on the basis of clinical history, physical examination, and radiographic findings (13).

For many years, a pleural fluid protein level of 3.0 g/dL was used to separate transudates from exudates, with exudative pleural effusions characterized by a protein level above 3.0 g/dL (14,15). Use of this one simple test led to the misclassification of approximately 10% of pleural effusions (14–16). Light et al. subsequently demonstrated that with the use of simultaneously obtained serum and pleural fluid protein and lactic acid dehydrogenase (LDH) values, 99% of pleural effusions could be correctly classified as either transudates or exudates (16). Exudative pleural effusions meet at least one of the following criteria, whereas transudative pleural effusions meet none (Light's criteria):

1. Pleural fluid protein divided by serum protein greater than 0.5
2. Pleural fluid LDH divided by serum LDH greater than 0.6
3. Pleural fluid LDH greater than two thirds of the upper limit of normal serum LDH

Subsequent studies have demonstrated that Light's criteria classify virtually all exudates correctly but misclassify about 25% of transudates as exudates.

In recent years, other tests have been proposed for the separation of transudates from exudates. The tests that have been proposed to indicate a pleural exudate have included a pleural fluid cholesterol greater than 60 mg/dL (17,18), a pleural fluid cholesterol greater than 45 mg/dL (19), a gradient of less than 1.2 g/dL for the difference in the pleural fluid and serum albumin level (20), a pleural fluid-to-serum bilirubin ratio above 0.6 (21), a high pleural fluid viscosity (22), a high level of oxidative stress markers (23), soluble leukocyte selectin (24), cytokines (25), uric acid (26), and a pleural fluid-to-serum cholinesterase ratio above 0.23 (27).

Two subsequent reports (28,29) have compared Light's criteria with some of the other proposed tests and have concluded that Light's criteria best separate exudates and transudates. In the study of Romero et al. (28) of 297 patients including 44 transudates and 253 exudates, Light's criteria were superior to cholesterol measurement in making the distinction. In this study with Light's criteria, 98% of the exudates and 77% of the transudates were correctly classified (28). In a subsequent study of 393 patients including 123 with transudates and 270 with exudates from South Africa (29), Light's criteria were found to be superior to the serum effusion albumin gradient, the effusion cholesterol concentration, and the pleura fluid-to-serum bilirubin ratio (29). Again in this study, Light's criteria identified 98% of the exudates correctly, but they were less accurate in identifying transudates, misclassifying 19 of 112 (17%) (29). Two additional studies (30,31) have come to similar conclusions. It is unlikely that the pleural fluid cholesterol measurement will provide additional information to the ratio of the pleural fluid to the serum protein because the pleural fluid cholesterol level can be accurately predicted from the serum cholesterol and the ratio of the pleural fluid to the serum protein level (32).

The number of false positives and false negatives with any test depends upon the cutoff level chosen for the identification of an exudate. If a high cutoff level is chosen, all transudates will be identified correctly, whereas if a low level is chosen, all exudates will be identified correctly. Light et al. originally developed Light's criteria with the goal to identify all exudates correctly, and the criteria are remarkably effective in achieving this goal.

An alternative approach is to select the cutoff level that will correctly identify the highest percentage of patients. Using this approach, Heffner et al. (33) analyzed the data from eight studies with a total of 1,448 patients and concluded that the best cutoff levels for the different pleural fluid tests were as follows: protein ratio 0.5, pleural fluid LDH 0.45 of the upper limits of the normal for serum, LDH ratio 0.45.

As demonstrated in the preceding text, Light's criteria identify approximately 25% of transudative effusions as exudates. This mislabeling occurs most commonly when patients with CHF are treated with diuretics before thoracentesis is performed (34). These mislabeled transudates barely meet the exudative criteria. The protein ratio is less than 0.65, the LDH ratio is less than 1.0 and the level of the LDH is less than the upper limit of normal. How can these mislabeled transudates be identified? One possible means is to examine the gradient between the serum and the pleural fluid protein levels. If this gradient is greater than 3.1 g/dL, one can presume that the fluid is actually a transudate (34). In a previous edition of this book, it was recommended that an albumin gradient of 1.2 g/dL (29) rather than the protein gradient of 3.1 g/dL be used. Bielsa et al. (35) reported that the albumin gradient identified more of these effusions correctly than did the protein gradient. It is suggested that the protein gradient first be examined because it is already available from Light's criteria. If the protein gradient is not definitive, then one may use the albumin gradient or the NT-pro-BNP.

In the discussion in the preceding text, pleural effusions have been dichotomized into transudates or exudates on the basis of a single cutoff point. An alternative approach is to use likelihood ratios for identifying whether a pleural fluid is a transudate or an exudate (36,37). The idea behind this approach is that the higher a value, for example, the pleural fluid LDH, the more likely the effusion is to be an exudate and the lower the value, the less likely the effusion is to be an exudate. Heffner et al. have derived multilevel (36) and continuous (37) likelihood ratios for the usual biochemical tests used to differentiate transudates and exudates. When these likelihood ratios are used in conjunction with pretest probabilities using Bayes' theorem, posttest probabilities can be derived (37). Difficulties in using this approach occur because the pretest probabilities vary significantly from physician to physician, and most physicians do not understand the mathematics involved. This approach does emphasize that it is important to take into consideration the absolute value of the measurements. Very high or very low values are almost always indicative of exudates and transudates, respectively, whereas values near the cutoff levels can be associated with either transudates or exudates.

The following approach is recommended for determining whether a pleural effusion is a transudate or an exudate. First assess the fluid with Light's criteria. The higher the value for the protein ratio, the LDH ratio, and the absolute value of the LDH, the more likely the fluid is an exudate. If the fluid meets the criteria for a transudative effusion, it is a transudate. If the fluid meets the criteria for an exudative effusion by only a small margin and the clinical picture is compatible with a transudative effusion, measure the protein gradient between the serum and pleural fluid. If this value is greater than 3.1 g/dL, then the fluid can be relabeled a transudate. An alternative approach is to measure the brain natriuretic peptide (BNP) level in the pleural fluid or the serum. If this is greater than 1,500 pg/mL, the diagnosis of CHF is established (see the discussion on NT-pro-BNP later in this chapter).

Specific Gravity

The specific gravity of the pleural fluid as measured with a hydrometer was used in the past to separate transudates from exudates (38) because it was a simple and rapid method of estimating the protein content of the fluid. A specific gravity of 1.015 corresponds to a protein content of 3 g/dL, and this value was used to separate transudates from exudates (38). At the present time, the specific gravity of pleural fluid is usually measured with a refractometer rather than a hydrometer. Unfortunately, the scale on the commercially available refractometers is calibrated for the specific gravity of urine rather than for pleural fluid. A reading of 1.020 on the urine specific gravity scale corresponds to a pleural fluid protein level of 3 g/dL. However, there is a scale on the same refractometer for protein levels that is valid for pleural fluid. Because the only reason to measure specific gravity is to estimate the protein level, and because the pleural fluid specific gravity measurement is extraneous and confusing, it should no longer be ordered (39). However, a rapid estimate of the pleural fluid protein content can be obtained at the patient's bedside with the protein scale on the refractometer (39).

Other Characteristics of Transudates

Most transudates are clear, straw colored, nonviscid, and odorless. It takes a pleural fluid RBC count of more than 10,000/mm^3 to give the pleural fluid a pinkish tinge. Approximately 15% have RBC counts above this level. Therefore, the discovery of blood-tinged pleural fluid does not mean that the fluid is not a transudate. Because RBCs contain a large amount of LDH, one might suppose that the LDH level in a blood-tinged or bloody transudative pleural effusion would be so elevated that it would meet the criteria for an exudative pleural effusion. Such does not appear to be the case, however. The LDH isoenzyme present in RBCs is LDH-1, and in one study of 23 patients with bloody pleural effusions (pleural fluid red cell counts greater than 100,000/mm^3), the fraction of LDH-1 in the pleural fluid was only slightly increased (40).

The pleural fluid white blood cell (WBC) count of most transudates is less than 1,000/mm^3, but approximately 20% have WBC counts that exceed 1,000/mm^3 (41). Pleural fluid WBC counts above 10,000/mm^3 are rare with transudative pleural effusions. The differential WBC count in transudative pleural effusions may be dominated by polymorphonuclear leukocytes, small lymphocytes, or other mononuclear cells. In a series of 47 transudative effusions, 6 (13%) had more than 50% polymorphonuclear leukocytes, 16 (34%) had predominantly small lymphocytes, 22 (47%) had predominantly other mononuclear cells, and 3 (6%) had no single predominant cell type (41). The pleural fluid glucose level is similar to the serum glucose level, and the pleural fluid amylase level is low (42). The pleural fluid pH with transudative pleural effusions is higher than the simultaneously obtained blood pH (43), probably because of active transport of bicarbonate from the blood into the pleural space (44).

Probrain Natriuretic Peptide (BNP)

The levels of NT-pro-BNP and BNP in the pleural fluid are useful in establishing that the etiology of the pleural effusion is CHF. When the ventricles are subjected to increased pressure or volume, these peptides are released into the circulation (45). The biologically active BNP and the larger amino terminal part NT-pro-BNP are released in equimolar amounts into the circulation (45). The serum levels of BNP are used to help establish the diagnosis of CHF. In clinical practice, levels above 500 pg/mL are considered diagnostic of CHF whereas levels below 100 pg/mL are thought to make the diagnosis of CHF unlikely (46).

Porcel et al. (47) first demonstrated that the pleural fluid levels of NT-pro-BNP are elevated in patients with heart failure. They measured NT-pro-BNP levels in 117 pleural fluid samples with the following diagnoses: CHF in 44 samples, malignancy in 25, tuberculous pleuritis in 20, hepatic hydrothorax in 10, and miscellaneous in 18. The mean NT-pro-BNP fluid level in the CHF patients (6,931 pg/mL) was significantly higher than the 551 pg/mL in the patients with hepatic hydrothorax and the 292 pg/mL in the patients with exudative pleural effusions (45). When a cutoff level of 1,500 pg/mL was used, the sensitivity was 91% and the specificity was 93% for the diagnosis of CHF. We have compared the pleural fluid NT-pro-BNP levels in 10 patients each with effusions due to CHF, pulmonary embolism, post-coronary artery bypass surgery, and malignancy (48). All the patients with CHF had NT-pro-BNP levels above 1,500 pg/mL, and none of the other patients had NT-pro-BNP levels this high (48).

It should be emphasized that the serum or pleural fluid BNP and NT-pro-BNP cannot be used interchangeably in the diagnosis of pleural effusions due to CHF (49). The BNP levels are only about 10% of the NT-pro-BNP levels. There is not a close correlation between the BNP levels and the NT-pro-BNP levels (r = 0.78) (50). Moreover, the diagnostic usefulness of the NT-pro-BNP in making the diagnosis of heart failure is superior to that of the BNP (50,51). The accuracy of NT-pro-BNP in making the diagnosis of pleural effusion due to heart failure was attested to in a meta-analysis of 10 studies with a total of 1,120 patients in which the pooled sensitivity and specificity were 94% and 94%, respectively (52).

The pleural fluid NT-pro-BNP is also superior to the BNP and the protein gradient in identifying patients with heart failure who meet Light's criteria for exudates (50). In one study of 20 patients with heart failure who met Light's criteria for exudates, 18 had NT-pro-BNP levels above 1,300, 16 had NT-pro-BNP levels above 1,500, but only 10 had serum pleural fluid protein gradient greater than 3.1 g/dL (50).

Other workers have demonstrated that there is a close relationship between the levels of NT-pro-BNP in the pleural fluid and serum. Han et al. (53) measured the NT-pro-BNP levels in 240 patients and reported that the correlation coefficient between the pleural and serum NT-pro BNP was 0.928. In a second study, Kolditz et al. (54) measured the serum and pleural fluid NT-pro-BNP levels in 93 patients including 25 with CHF. They confirmed the results of the above study in that the levels of serum and pleural fluid NT-pro-BNP were again closely correlated ($r^2 = 0.90$). From the latter two studies it appears that measurement of the pleural fluid NT-pro-BNP levels provides no additional information beyond the serum measurements.

GENERAL TESTS FOR DIFFERENTIATING CAUSES OF EXUDATES

Appearance of Fluid

The gross appearance of the pleural fluid frequently yields useful diagnostic information. The color, turbidity, viscosity, and odor should be described. Most transudative and many exudative pleural effusions are clear, straw colored, nonviscid, and odorless. Any deviations should be noted and investigated.

A reddish color indicates that blood is present, and a brownish tinge indicates that the blood has been present for a prolonged period. If the pleural fluid is blood tinged, the pleural fluid RBC count is between 5,000 and 10,000/mm^3. If the pleural fluid appears grossly bloody, a hematocrit should be obtained to determine whether the patient has a hemothorax (see Chapter 25).

On rare occasions, the pleural fluid can be black. Black pleural fluid has been reported with infection due to *Aspergillus niger*, infection due to *Rhizopus oryzae*, pigment laden macrophages following massive bleeding due to metastatic carcinoma (55) and melanoma (56).

Turbid pleural fluid can occur from either increased cellular content or increased lipid content. These two entities can be differentiated if the pleural fluid is centrifuged and the supernatant examined. If turbidity remains after centrifugation, it is in all probability due to increased lipid content, and the fluid should be sent for lipid analysis (see the discussion later in this chapter).

Alternately, if the supernatant is clear, the original turbidity was due to increased numbers of cells or other debris. The discovery of pleural fluid that looks like chocolate sauce or anchovy paste is suggestive of amebiasis with a hepatopleural fistula (57). This appearance is due to the presence of a mixture of blood, cytolyzed liver tissue, and small solid particles of liver parenchyma that have resisted dissolution.

A pleural fluid with a high viscosity is suggestive of malignant mesothelioma; the high viscosity is secondary to an elevated pleural fluid hyaluronic acid level. Of course, the fluid from a pyothorax is also viscid because of the large amounts of cells and debris in the fluid.

The odor of all pleura fluids should be noted. One can immediately establish two diagnoses by smelling the pleural fluid. A feculent odor indicates that the patient has a bacterial infection of the pleural space that is probably anaerobic. If the pleural fluid smells like urine, the patient probably has a urinothorax.

Red Blood Cell Count

Only 5,000 to 10,000 RBCs per mm^3 need be present to impart a red color to pleural fluid. If a pleural effusion has a total volume of 500 mL and the RBC count in the peripheral blood is 5 million/mm^3, a leak of only 1 mL blood into the pleural space will result in a blood-tinged pleural effusion. It is probably for this reason that the presence of blood-tinged or serosanguineous pleural fluid has little diagnostic significance. More than 15% of transudative and more than 40% of all types of exudative pleural fluids are blood tinged (41); that is, they have pleural fluid RBC counts between 5,000 and 100,000/mm^3.

Occasionally, pleural fluid obtained by diagnostic thoracentesis appears grossly bloody. In such cases, one can assume that the RBC count in the pleural fluid is above 100,000/mm^3. One should obtain a hematocrit on such pleural fluids to document the amount of blood in the pleural fluid. If the hematocrit of the pleural fluid is greater than 50% of the peripheral hematocrit, a hemothorax is present, and one should consider inserting a chest tube (see Chapter 25). Usually, the hematocrit of bloody pleural fluid is much lower than what one would expect from its gross appearance. If a pleural fluid hematocrit is not available, its estimate can be obtained by dividing the pleural fluid RBC count by 100,000.

The presence of bloody pleural fluid suggests one of three diagnoses, namely, malignant disease, trauma, or pulmonary embolization. In a series of 22 bloody pleural effusions that I observed on medical wards, 12 were due to malignant disease, 5 to pulmonary embolization, 2 to trauma, and 2 to pneumonia, and 1 was a transudative effusion secondary to cirrhosis (30). The traumatic origin of the pleural effusion may not be obvious, particularly when the patient is on a medical ward. The patient may have broken a rib while coughing or suffered trauma during an episode of inebriation that is not remembered.

At times, it is unclear whether blood in the pleural fluid resulted from or was present before the thoracentesis. If the blood is a result of the thoracentesis, the degree of red discoloration of the fluid frequently is not uniform throughout the course of aspiration. Examination of the fluid microscopically may also be useful. If the RBCs were present before the thoracentesis, the macrophages in the pleural fluid usually contain hemoglobin inclusions. Although the levels of D-dimer in the cerebrospinal fluid are useful in demonstrating whether blood in the fluid has a traumatic origin, pleural fluid levels of D-dimer are not useful in making this differentiation (58). Crenation of the RBC in the pleural fluid rarely occurs because the osmotic pressure of the pleural fluid is similar to that of serum.

White Blood Cell Count

Although WBC counts on the pleural fluid were traditionally performed manually, we have shown that the automated counters provide accurate pleural fluid WBC counts (59). The pleural fluid for cell counts and differentials should be collected in a test tube with an anticoagulant (59). If the pleural fluid is collected in plastic or glass tubes without an anticoagulant, the fluid may clot or the cells may clump, providing inaccurate cell counts and differentials (59).

The pleural fluid WBC count is of limited diagnostic use. Most transudates have WBC counts below 1,000/mm^3, whereas most exudates have WBC counts above 1,000/mm^3 (16). Pleural fluid WBC counts above 10,000/mm^3 are most commonly seen with parapneumonic effusions, but they are also seen with many other diseases (41), as shown in Table 7.1. I have seen pleural fluid WBC counts above 50,000/mm^3 with both pancreatic disease and pulmonary embolization. With grossly purulent pleural fluid, the pleural fluid WBC count is frequently much lower than what one would anticipate because debris, rather than cells, accounts for much of the turbidity.

Differential White Cell Count

Examination of a Wright's stain of pleural fluid is one of the most informative tests on pleural fluid. Because the pleural fluid WBC count is frequently less than 5,000/mm^3, it is useful to concentrate the cells before staining. This is easily accomplished by centrifuging approximately 10 mL of fluid and then resuspending the button of cells in approximately 0.5 mL of supernatant. After thorough mixing, slides that are similar to those for examining peripheral blood are made and stained in the usual way. Occasionally, large amounts of fibrinogen adhere to the cells. In such cases, resuspension in saline solution, followed by centrifugation, is indicated in order to evaluate cellular morphologic features. Automatic cell counters do not provide sufficiently accurate differential cell counts for clinical use (59,60), presumably because of the high number of mesothelial, lymphoid, and tumors cells in pleural fluid.

TABLE 7.1 ■ **Etiology of 25 Effusions Containing More than 10,000 WBCs/mm^3**

Diagnosis	Number with >10,000 WBCs/mm^3	Total Number of Effusions	Percentage Having >10,000 WBCs/mm^3
Parapneumonic effusion	13	26	50
Malignant disease	3	43	7
Pulmonary embolization	3	8	37
Tuberculosis	2	14	14
Pancreatitis	2	5	40
Postmyocardial infarction Syndrome	1	3	33
Systemic lupus erythematosus	1	1	100

WBC, white blood cell.

Although most laboratories divide pleural fluid WBCs into polymorphonuclear leukocytes and mononuclear cells, I prefer to divide them into four categories—polymorphonuclear leukocytes, lymphocytes, other mononuclear cells, and eosinophils—because of the diagnostic significance of small lymphocytes (see the discussion on lymphocytes later in this chapter). The mononuclear cells include mesothelial cells, macrophages, plasma cells, and malignant cells. Excellent color plates demonstrating the morphologic and staining characteristics of the different cells in pleural effusions are contained in the monograph by Spriggs and Boddington (61).

Neutrophils

Because neutrophils are the cellular component of the acute inflammatory response, they predominate in pleural fluid resulting from acute inflammation as with pneumonia, pancreatitis, pulmonary embolization, subphrenic abscess, and early tuberculosis. Although more than 10% of transudative pleural effusions contain predominantly neutrophils, pleural fluid neutrophilia in transudates has no clinical significance (41). The significance of neutrophils in an exudative pleural effusion is that they indicate acute inflammation of the pleural surface.

Interleukin 8 (IL-8) appears to be one of the primary chemotaxins for neutrophils in the pleural space (62,63). The number of neutrophils in pleural fluid is correlated with the IL-8 level, and empyemas have the highest levels of IL-8. The addition of IL-8 neutralizing serum decreases the chemotactic activity for neutrophils in empyema fluids (63). The cellular source of IL-8 is unknown (62).

Examination of the pleural fluid neutrophils in patients with parapneumonic effusions is useful in identifying those that are infected. If pleural infection is present, the neutrophils undergo a characteristic degeneration. The nucleus becomes blurred and is no longer stained purple. The cytoplasm shows toxic granulation initially. Subsequently, the neutrophilic granules become indistinct and are then lost. Finally, only a smear cell remains (61).

Eosinophils

Approximately 7% of pleural fluids are characterized by pleural fluid eosinophilia (>10%) (64,65). Most clinicians believe that significant numbers of eosinophils (>10%) in pleural fluid should be a clue to the origin of the pleural effusion. In most instances, the pleural fluid eosinophilia is due to either air or blood in the pleural space and therefore does not contribute any diagnostic information in these situations. Charcot–Leyden crystals (66), as well as Curschmann's spirals (67), are occasionally found in the pleural fluid of patients with pleural eosinophilia. Their presence appears to have no diagnostic significance.

The factors responsible for pleural fluid eosinophilia have been intensively studied, but there is still no unifying concept and it is likely that multiple factors are involved. In animals, it has been shown that the intrapleural injection of stem cell factor (68), platelet-activating factor (PAF) (69), endotoxin (70,71), bradykinin (72), and leukotriene B_4 (69) all result in pleural fluid eosinophilia. The eosinophil influx is inhibited in some but not all situations by monoclonal antibodies (MAb) directed against IL-5.

In humans, pleural fluid from patients with eosinophilic pleural effusions stimulates bone marrow cells to form colonies of eosinophils (73,74). In addition, when eosinophils are incubated in the presence of eosinophilic pleural fluid, their survival is prolonged (73). Peripheral blood from patients with eosinophilic pleural effusions does not stimulate the bone marrow to form eosinophil colonies and does not prolong survival of eosinophils. The factor responsible for the increased colony-forming activity and the increased survival appears to be IL-5 (73,74), although IL-3 and granulocyte or macrophage colony-stimulating factor (GM-CSF) may also play a role (73). In patients with eosinophilic pleural effusions, the pleural fluid level of IL-5 is significantly correlated with the pleural fluid eosinophil count (r = 0.55) and the pleural fluid eosinophil percentage (r = 0.54) (75). Both bloody and nonbloody eosinophilic effusions have high levels of IL-5 (75). In patients with eosinophilic pleural effusions, the pleural fluid levels of vascular cell adhesion molecule (VCAM)-1 and eotaxin-3 are also significantly correlated with the eosinophil count and percentage of eosinophils in the pleural fluid (76). VCAM-1 is an adhesion molecule that is expressed on the endothelial surface and interacts with the β1-integrins expressed on the eosinophil surface to facilitate eosinophil tissue migration (76).

The source of the IL-5 appears to be the $CD4^+$ lymphocyte in the pleural fluid (74), but the eosinophils in the pleural fluid may themselves also produce IL-5 (73). The source of the eosinophils in eosinophilic pleural effusions appears to be the bone marrow; no progenitor cells are present in the pleural space. It is not known what stimulates the $CD4^+$ lymphocytes to produce the IL-5. However, it probably results from another cytokine because the intrapleural injection of IL-2 into malignant pleural effusions results in an eosinophilic pleural effusion with a high level of IL-5 (77). There are factors other than IL-5 that recruit eosinophils to the pleural space. Antibodies to IL-5 eliminate the eosinophilic influx to an allergen but not to endotoxin in the mouse (78). However, antibodies to gamma delta lymphocytes eliminate the eosinophilic response to endotoxin (71).

The most common cause of pleural fluid eosinophilia is air in the pleural space. In a series of 127 cases with more than 20% eosinophils in the pleural fluid, 81 (64%) were thought to have pleural fluid eosinophilia secondary to air in the pleural space (61). In a review of 343 pleural effusions with greater than 10% eosinophils, 95 (28%) had air in the pleural space (79). It is likely that the pleural fluid eosinophilia in many other patients in this series was also due to the introduction of air into the pleural space during a prior thoracentesis. On numerous occasions over the past three decades, I have seen patients who had no pleural fluid eosinophilia at the initial thoracentesis but who had many eosinophils at a subsequent thoracentesis. In each case, a small pneumothorax resulted from the initial thoracentesis. When patients with spontaneous pneumothorax undergo thoracotomy, a reactive eosinophilic pleuritis frequently exists in the resected parietal pleura (80). It should be noted, however, that in two recent series (64,81) the prevalence of eosinophilic pleural effusion did not increase after a thoracentesis or a pleural biopsy.

The mechanism responsible for the pleural fluid eosinophilia in response to air in the pleural space is unknown but is probably related to IL-5. Smit et al. (82) measured the percentage of eosinophils and the levels of IL-5 in 23 patients with pneumothorax and pleural fluid. They found that IL-5 level and the eosinophil concentration in the pleural fluid were highly correlated ($r = 0.84$) and that the eosinophil percentage tended to increase with time, with a mean of less than 5% in the first 24 hours, 20% at days 1 to 3, 40% at days 4 to 7, and 50% after day 7 (82). There was no relationship between the PAF level or the monocyte chemotactic protein-1 levels and the eosinophils. In these fluids, IL-8 was not detectable. When air is injected into the pleural space of a mouse, pleural fluid eosinophilia occurs within 30 minutes and peaks at 48 hours (83).

The second most common cause of pleural fluid eosinophilia is blood in the pleural space. Following traumatic hemothorax, pleural fluid eosinophils do not usually become numerous until the second week (61). There is frequently an associated peripheral blood eosinophilia that does not disappear until the pleural effusion is completely resolved (84). The pleural effusions associated with pulmonary embolization are frequently bloody and contain numerous eosinophils (85). Bloody pleural fluids due to malignant disease are not usually characterized by eosinophilia (41,61). In a study conducted by my colleagues and me of bloody pleural effusions, none of the 11 cases of malignant pleural effusions with pleural fluid RBC counts greater than 100,000/mm^3 had more than 10% eosinophils (41). Patients with pleural effusions postcoronary artery bypass graft (CABG) surgery, frequently have bloody pleural effusions in the first few weeks after surgery. In these effusions, the pleural fluid IL-5 level is higher than the corresponding serum level and there is a significant correlation between the pleural fluid and serum IL-5 level (86). Moreover, the pleural fluid IL-5 levels are significantly correlated with the pleural fluid eosinophil counts (86). In addition, the pleural fluid eotaxin-3 levels are significantly higher than the serum levels and the pleural fluid eotaxin-3 levels significantly correlate with the pleural fluid eosinophil count (86).

If the patient has neither blood nor air in their pleural space, what is the significance of an eosinophilic pleural effusion? Kalomenidis and Light (87) reviewed the etiology of 392 cases of eosinophilic pleural effusions when cases associated with pleural air and/or blood were excluded. They reported that the most common diagnosis was idiopathic (39.8%), followed by malignancy (17%), parapneumonic (12.5%), transudates (7.9%), tuberculosis (5.6%), pulmonary embolism (4.3%), and others (12.8%). In a recent study of 135 patients with eosinophilic pleural effusions from a single institution, the following diseases were responsible: malignancy 34.8%, infections 19.2%, unknown 14.1%, posttraumatic 8.9%, and miscellaneous 23.0%. (64). The incidence of malignancy was significantly higher in patients with a lower (<40%) pleural fluid eosinophil percentage (64).

If neither air nor blood is present in the pleural space, several unusual diagnoses should be considered. Pleural eosinophilia is common in patients with asbestos-related pleural effusions. In a review of eosinophilic pleural effusions (79), 15 of 29 (52%) asbestos-related pleural effusions had more than 10% eosinophils in the pleural fluid. Patients with eosinophilic pneumonia frequently have pleural effusions and the mean eosinophil percentage in the pleural fluid was 38% in one study (88). The pleural effusions secondary to drug reactions are frequently eosinophilic.

Offending drugs include dantrolene, bromocriptine, and nitrofurantoin (see Chapter 22). Pleural effusions secondary to parasitic diseases (see Chapter 15) such as paragonimiasis (68), hydatid disease (89), amebiasis (61), or ascariasis (61) frequently contain a large percentage of eosinophils. Lastly, the pleural effusion associated with the Churg-Strauss syndrome (see Chapter 21) is eosinophilic (90).

If none of the foregoing rare diseases is causing the pleural effusion, the following statements are pertinent to patients with eosinophilic pleural effusions. If the patient has pneumonia and pleural effusion, the presence of pleural fluid eosinophilia is a good prognostic sign because such an effusion rarely becomes infected. The origin of approximately 40% of eosinophilic effusions is not established, and these effusions resolve spontaneously.

Basophils

Basophilic pleural effusions are distinctly uncommon. I have not seen a pleural effusion that contained more than 2% basophils. A few basophils are usually present in pleural effusions with eosinophils. Basophil counts greater than 10% are most common with leukemic pleural involvement (61). Basophil counts greater than 5% are most common with pneumothorax, pneumonia, and transudates (91).

Lymphocytes

The discovery that more than 50% of the WBCs in an exudative pleural effusion are small lymphocytes is diagnostically important because it means that the patient probably has a malignant disease, tuberculous pleuritis, or a pleural effusion after CABG surgery. In two series (41,92) studied before the advent of CABG surgery, 96 of 211 exudative pleural effusions had more than 50% small lymphocytes. Of these 96 effusions, 90 (94%) were due to tuberculosis or malignant disease.

When the foregoing series are analyzed, almost all of the effusions secondary to tuberculosis (43 of 46), but only two thirds of the effusions secondary to malignant disease (47 of 70), had predominantly small lymphocytes. In one series of 26 patients with chronic pleural effusions post-CABG, the mean percentage of lymphocytes in the pleural fluid was 61% (93). Approximately one third of transudative pleural effusions contain predominantly small lymphocytes (41), and a lymphocytic transudative effusion is not an indication for pleural biopsy.

Several papers have assessed the diagnostic utility of separating pleural lymphocytes into T and B lymphocytes (94–97). In general, this separation has not been useful diagnostically. With most disease states, the pleural fluid contains a higher percentage of T lymphocytes (70%), a lower percentage of B lymphocytes (10%), and a higher percentage of null cells (20%) than the corresponding peripheral blood (94,95). The partitioning of lymphocytes may be useful, however, when chronic lymphatic leukemia or lymphoma is suspected. In a report of four such patients, all had more than 80% B lymphocytes in their pleural fluid (96).

The development of MAb has permitted a further subdivision of T lymphocytes. In comparison to peripheral blood, in pleural fluid the ratio of the helper and inducer cells ($CD4^+$) to the suppressor and cytotoxic cells ($CD8^+$) is higher, regardless of the etiology of the pleural effusion (98–100). Therefore, this subdivision is not useful diagnostically.

Natural killer (NK) cells are lymphocytes derived from an unimmunized host that lyse certain tumor cell lines and virus-infected cells. In general, the percentage of T lymphocytes in pleural fluid that are identified as NK cells is approximately the same (~15%) as in the peripheral blood (101). There are two types of NK cells, $CD56^{bright}$ and $CD56^{dim}$. In the pleural fluid, there is a much higher percentage of $CD56^{bright}$ cells than $C56^{dull}$ cells (101). The $CD56^{dim}$ subset has cytotoxic activity that is superior to that of the $CD56^{bright}$ subset. However, there is a discrepancy between the number of NK positive cells and the NK activity of the cells when patients with tuberculosis are compared with patients with malignancy. Although the number of NK cells is comparable in the two populations, there is much more NK activity in the tuberculous pleural effusions (102). Differences in the NK subset percentages may explain the variation in the NK activity (103).

Mesothelial Cells

Mesothelial cells line the pleural cavities. They frequently become dislodged from the pleural surfaces and are present in the small amount of normal pleural fluid (104). These cells are usually 12 to 30 μm in diameter, but multinucleated forms may have diameters up to 75 μm. Their cytoplasm is light blue (Fig. 7.1A) and often contains a few vacuoles. The nucleus is large (9–22 μm) and stains purplish with a uniform appearance. The nucleus usually contains one to three bright blue nucleoli (61). Mesothelial cells are discussed in more detail in Chapter 1.

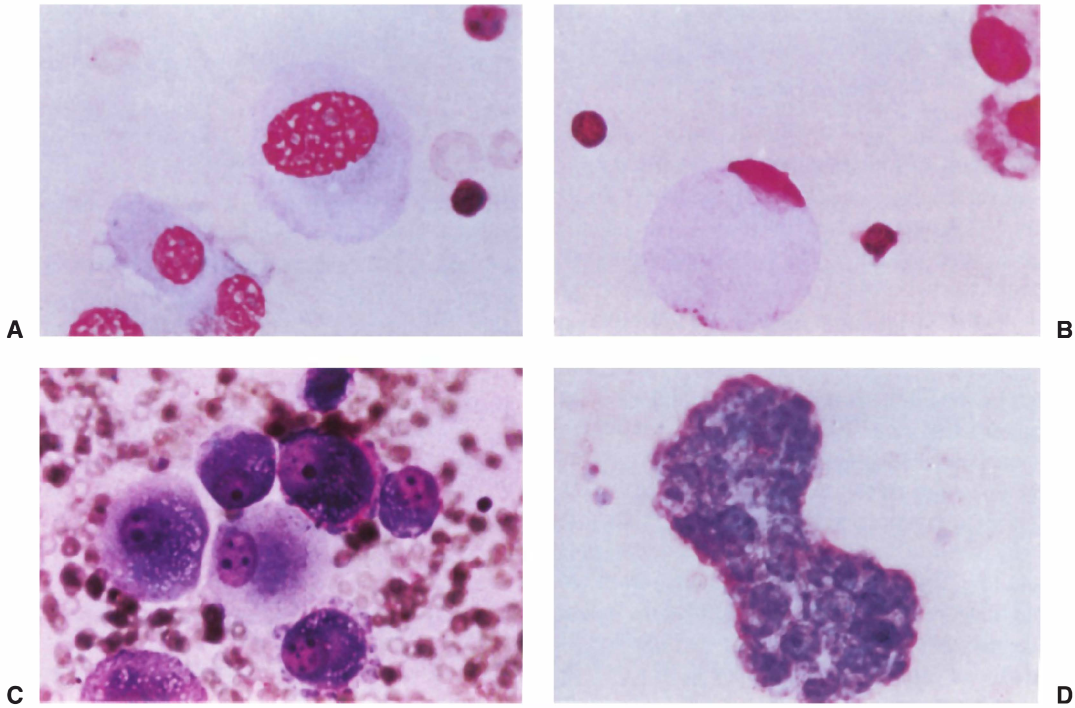

FIGURE 7.1 ■ **A**: Mesothelial cell. Note that the mesothelial cell is large in comparison to the lymphocyte and has light blue cytoplasm and light blue nucleoli of varying shapes and sizes. **B**: Signet-ring cell. This macrophage, which has engorged itself with pleural debris, is not a malignant cell. **C**: Malignant cells. Several large cells are similar but vary in size. Note the large, dark nucleoli, which are so different from the nucleoli in the mesothelial cell. **D**: Clump of malignant cells from the pleural fluid of a patient with metastatic adenocarcinoma.

Mesothelial cells are significant for two reasons. First, their presence or absence is often useful diagnostically because these cells are uncommon in tuberculous effusions. Spriggs and Boddington (61) analyzed 65 tuberculous effusions and found that only one effusion had more than a single mesothelial cell per 1,000 cells (61). Light et al. have confirmed the paucity of mesothelial cells in tuberculous pleural effusions (41), as have Yam (92) and Hurwitz et al. (105). The exception to this observation is the patient with acquired immunodeficiency syndrome (AIDS). Patients with AIDS having a low $CD4^{+}$ count who have tuberculous pleuritis may have numerous mesothelial cells in their pleural fluid (106). The lack of mesothelial cells is not diagnostic of tuberculosis, however. It simply indicates that the pleural surfaces have become extensively involved by the disease process so that the mesothelial cells cannot enter the pleural space. The absence of mesothelial cells is common with complicated parapneumonic effusions and with other conditions in which the pleura becomes coated with fibrin. It is also common with malignant effusions after sclerosing agents have been injected to effect a pleurodesis. Second, mesothelial cells, particularly in their activated form, may be confused with malignant cells. Frequently, an experienced pathologist is required to make the differentiation. Immunohistochemistry is useful in making this distinction (see discussion later in this chapter).

Macrophages

In general, the presence of macrophages in pleural fluid is of limited diagnostic use. By definition, macrophages are cells that store vital dyes. It appears that the origin of the pleural fluid macrophages can be either the circulating monocyte or the mesothelial cell (107). Macrophages vary in diameter from 15 to 50 μm and have irregular nuclei. Their cytoplasm is gray, cloudy, and full of vacuoles. At times, the macrophage may become engorged with debris, taking on the appearance of a "signet-ring" cell, with the nucleus flattened against the side of the cell (Fig. 7.1B). It is important not to confuse these cells with malignant cells. During phagocytosis, macrophages may engulf polymorphonuclear leukocytes or RBCs. These cells may be evident within the macrophage in various stages of digestion. If RBCs have been ingested, the iron pigment is retained as dark blue or brown staining material (61). It is important not to confuse macrophages with mesothelial cells because macrophages are sometimes present in tuberculous pleural effusions (61).

Macrophages in the pleural fluid can definitively be identified using the MAb CD68 (108) or CD14 and using either flow cytometry or immunohistochemistry (109). Most pleural fluids have more than 10% macrophages (109). The macrophages in the pleural fluid, in addition to their phagocytic function, can also release IL-1 and tumor necrosis factor (TNF) (108). The macrophages in the pleural fluid are also efficient accessory cells for T-cell proliferation (108). Mice depleted of macrophages have a dramatic reduction in the neutrophil influx in response to intrapleural carrageenan or *Staphylococcus aureus* (110).

Plasma Cells

The presence of numerous plasma cells in the pleural fluid suggests multiple myeloma. Smaller numbers of plasma cells are not of any particular diagnostic importance. These cells are of the lymphoid series and produce immunoglobulins. Morphologically, they are larger than small lymphocytes and have an eccentric nucleus and deeply staining basophilic cytoplasm with a clear area at the cell center (Golgi zone) (61). Mature forms have well-defined nuclear chromatin blocks. In a series of 18 effusions with more than 5% plasma cells, 4 were due to malignant disease, 3 due to tuberculosis, 3 due to CHF, 3 due to pulmonary embolization, 2 due to pneumonia, 1 due to sepsis, and 2 were of undetermined origin (61).

Dendritic Cells

Dendritic cells are human leukocyte antigen (HLA) DR–positive accessory cells that play a critical role in the development of cell-mediated immune reactions. Dendritic cells have been identified in the pleural fluid (111). It is unknown whether the dendritic cells in pleural fluid are resident in the pleural space or reach the pleural space from the blood. Light microscopy reveals that dendritic cells from pleural fluid are intermediate in size between lymphocytes and macrophages, do not contain intracytoplasmic inclusions, and have an eccentric nucleus (111).

PROTEIN MEASUREMENTS

The pleural fluid protein levels are generally higher in exudative pleural effusions than in transudative pleural effusions, and this observation serves as a basis for separating transudates from exudates (see the foregoing discussion on this differentiation in this chapter). Pleural fluid protein levels are not useful in separating the various types of exudative effusions, however, because the protein level in most exudates is elevated to a comparable degree (Fig. 7.2). At times, a pleural fluid meets the exudative criteria with its LDH, but not with its protein level. Such exudative pleural effusions are almost always parapneumonic effusions or are secondary to malignant pleural disease (16).

Simultaneous electrophoretic studies of serum and pleural fluid demonstrate that the pattern in the pleural fluid is essentially an image of that in the serum, except that proportionately more albumin is present in the pleural fluid (112,113). Along the same lines, the ratio of the pleural fluid to the serum IgG, IgA, and IgM is always below unity and appears to have no diagnostic value (113). The ratio of the concentration of these proteins is inversely related to their molecular weight (113). The immunoglobulin measurement that may be diagnostically useful is that of IgE. Yokogawa et al. (114) measured the pleural fluid and serum IgE levels in five patients with paragonimiasis. In all five patients, the pleural fluid IgE level was above 4,000 IU and exceeded the serum level. A subsequent report, however, measured the pleural fluid and serum IgE levels in seven patients with eosinophilic pleural effusions and seven with noneosinophilic pleural effusions. The pleural fluid levels and the ratio of the pleural fluid and serum IgE concentration were comparable in the two groups, and it was concluded that pleural fluid IgE was the result of passive diffusion from the serum (115). None of the patients in this latter series, however, had paragonimiasis.

GLUCOSE MEASUREMENT

Measurement of the pleural fluid glucose level is useful in the differential diagnosis of exudative pleural effusions because a low pleural fluid glucose level (<60 mg/dL) indicates that the patient probably has one of four disorders, namely, a parapneumonic effusion, malignant disease, rheumatoid disease, or tuberculous pleuritis. Other rare causes of a low glucose pleural effusion include paragonimiasis, hemothorax, Churg-Strauss syndrome, and, occasionally, lupus pleuritis. The pleural fluid glucose level of all transudates and of most exudates parallels that of the serum.

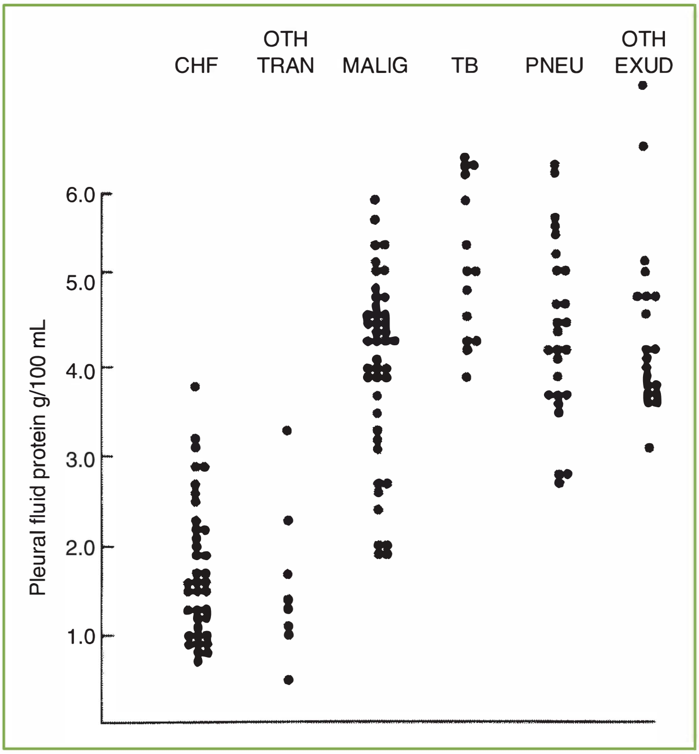

FIGURE 7.2 ■ Pleural fluid protein levels in effusion secondary to congestive heart failure (CHF), other transudates (OTH TRAN), malignant disease (MALIG), tuberculosis (TB), pneumonia (PNEU), and other exudates (OTH EXUD). Each point represents one pleural fluid. Note that the distribution of protein levels for all categories of exudative pleural effusions is similar. *(From Light RW, MacGregor MI, Luchsinger PC, et al. Pleural effusions: the diagnostic separation of transudates and exudates.* Ann Intern Med. *1972;77:507–513, with permission.)*

In my experience, it is not necessary to obtain pleural fluid glucose levels with the patient fasting or to take the serum glucose level into consideration when evaluating the pleural fluid glucose level.

The pleural fluid glucose level is low in some patients with parapneumonic effusions or empyema (116,117). The lower the pleural fluid glucose level, the more likely that one is dealing with a complicated parapneumonic pleural effusion. If the pleural fluid is thick and purulent, the pleural fluid glucose level is frequently close to zero (116). Even in more serous fluid, the glucose level may be reduced. The presence of a low pleural fluid glucose is a poor prognostic sign in patients with parapneumonic effusion and serves as an indicator that more aggressive therapy such as tube thoracostomy or thoracoscopy with the breakdown of loculations is necessary (118).

Approximately 15% to 25% of patients with malignant pleural effusions have pleural fluid glucose levels below 60 mg/dL (42,119,120) and the level may be less than 10 mg/dL. Patients with malignant pleural effusions and a low glucose level have a greater tumor burden in their pleural space than do those with normal pleural fluid glucose levels. In one report of 77 patients with malignant pleural effusion who underwent thoracoscopic examination (120), the extent of the tumor at thoracoscopy was significantly higher in those 16 patients in whom the pleural fluid glucose was less than 60 mg/dL. In addition, patients with a low pleural fluid glucose are more likely to have positive pleural fluid cytology and a positive pleural biopsy (121), are less likely to have a good result from chemical pleurodesis (120,122), and have a shortened life expectancy (122,123).

The pleural fluid glucose level is also reduced in some patients with tuberculous pleuritis. Indeed, early reports indicated that low pleural fluid glucose levels were seen only with tuberculous pleural effusions (124,125). Subsequent studies (41,116,119,126), however, revealed that low pleural fluid glucose levels also occurred with malignant and rheumatoid disease and parapneumonic effusion.

The distribution of pleural fluid glucose levels for tuberculous and malignant pleural effusions is, in fact, similar (42). Most patients with tuberculous pleuritis have a pleural fluid glucose level above 80 mg/dL (42). Accordingly, a low pleural fluid glucose level is compatible with the diagnosis of tuberculous pleuritis, but it is not necessary for the diagnosis.

Pleural effusions due to rheumatoid disease (see Chapter 21) classically have a low pleural fluid glucose level. Carr and Power (126) first reported that rheumatoid pleural effusions had a low pleural fluid glucose level. In a subsequent review of 76 cases of rheumatoid pleural effusions (127), 42% had pleural fluid glucose levels below 10 mg/dL, and 78% had levels below 30 mg/dL. The explanation for the low pleural fluid glucose level in this condition appears to be a selective block to the entry of glucose into the pleural effusion (128). The pleural fluid glucose level in effusions secondary to LE is usually normal. In one report of nine patients, the pleural fluid glucose; the pleural fluid glucose level was below 50 mg/dL in 2 of 14 (14%) patients with lupus pleuritis.

AMYLASE DETERMINATION

Pleural fluid amylase determinations are useful in the differential diagnosis of exudative pleural effusions because a pleural fluid amylase level above the upper normal limits for serum indicates that the patient has one of three problems: pancreatic disease, malignant tumor, or esophageal rupture (42). However, it is not cost effective to obtain an amylase measurement on every undiagnosed pleural fluid; rather, pleural fluid amylase levels should be determined only when esophageal rupture or pancreatic disease is suspected (131).

Approximately 50% of patients with inflammatory pancreatic disease have an accompanying pleural effusion (132). In such patients, the pleural fluid amylase level is usually raised well above the normal upper limits for serum and is also higher than the simultaneously sampled serum (42,133). On rare occasions, the pleural fluid amylase level is normal at the time of the original thoracentesis, only to become elevated at the time of a subsequent thoracentesis. In some patients with acute pancreatitis with pleural effusion, the chest symptoms of pleuritic chest pain and dyspnea may overshadow the abdominal symptoms. In such instances, an elevated pleural fluid amylase level may be the first hint of a pancreatic problem (42).

Patients with chronic pancreatic disease may also present with a pleural effusion with a high amylase content (134). The effusion results when a sinus tract connects the pancreatic pseudocyst and the pleural space. The patients typically appear chronically ill without abdominal symptoms and appear to have cancer. If a pleural fluid amylase level is not measured, the correct diagnosis may never be established. The key to this diagnosis is a markedly elevated (>1,000 U/L) pleural fluid amylase level (135).

The pleural fluid amylase level is elevated in approximately 10% of malignant pleural effusions (42,135). The serum amylase level is also elevated in approximately 50% of patients with malignant pleural effusions and an elevated pleural fluid amylase. The pleural fluid amylase level in malignant pleural effusions is usually only minimally to moderately elevated, in contrast to the marked elevations with pancreatitis or esophageal rupture. The primary site of the tumor in patients with neoplastic pleural effusions and elevated pleural fluid amylase levels is usually not the pancreas (42,136). Because the amylase in malignant pleural effusions is of the salivary type (137), amylase isoenzyme determinations are useful in distinguishing between malignant and pancreas-related pleural effusions with high amylase levels.

The pleural fluid amylase level is also elevated with esophageal rupture (42,138). The origin of the amylase with esophageal rupture has been shown to be the salivary gland rather than the pancreas (139). With the tear in the esophagus, the swallowed saliva with its high amylase content passes into the pleural space. Because the early diagnosis of esophageal perforation is imperative, owing to the high mortality rate without rapid operative intervention, the pleural fluid amylase determination should be performed promptly when this diagnosis is suspected. In animal experiments, the pleural fluid amylase concentration is elevated within 2 hours of esophageal rupture (140).

LACTIC ACID DEHYDROGENASE MEASUREMENT

The pleural fluid LDH level is used to separate transudates from exudates (see discussion earlier in this chapter). Most patients who meet the criteria for exudative pleural effusions with LDH but not with protein levels have either parapneumonic effusions or malignant pleural disease. Although initial reports suggested that the pleural fluid LDH level was increased only in patients with malignant pleural disease (141), subsequent reports demonstrated that the pleural fluid LDH was elevated in most exudative effusions regardless of origin (Fig. 7.3), and therefore, this determination is of no use in the differential diagnosis of exudative pleural effusions (16).

Nevertheless, every time that I perform a thoracentesis, I obtain a pleural fluid LDH level. This is because the level of the pleural fluid LDH is a reliable indicator of the degree of pleural inflammation; the higher the LDH, the more inflamed the pleural surfaces. Serial measurement of the pleural fluid LDH levels is informative when one is dealing with a patient with an undiagnosed pleural effusion. If with repeated thoracenteses the pleural fluid LDH level becomes progressively

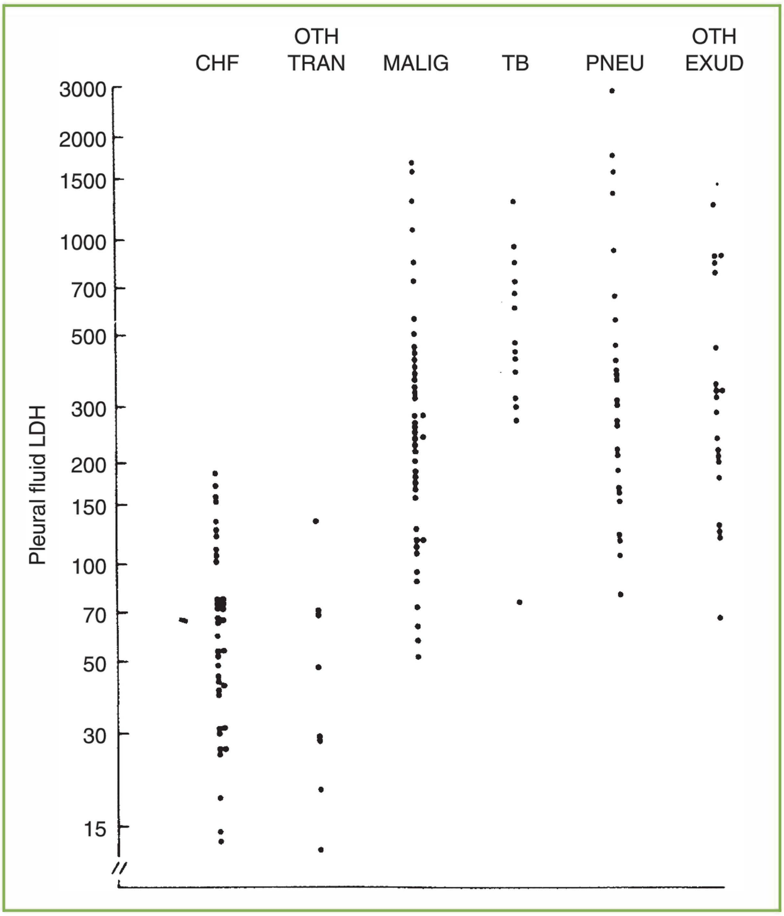

FIGURE 7.3 ■ Pleural fluid lactic dehydrogenase (LDH) levels. Note the similar distributions of the LDH levels for all categories of exudative pleural effusions. CHF, congestive heart failure; OTH TRAN, other transudates; MALIG, malignant disease; TB, tuberculosis; PNEU, pneumonia; OTH EXUD, other exudates. *(From Light RW, MacGregor MI, Luchsinger PC, et al. Pleural effusions: the diagnostic separation of transudates and exudates.* Ann Intern Med. *1972;77:507–513, with permission.)*

higher, the degree of inflammation in the pleural space is increasing and one should be aggressive in pursuing a diagnosis. Alternatively, if the pleural fluid LDH level decreases with time, the process is resolving and one need not be as aggressive in the approach to the patient. It should be noted that needle biopsy of the pleura results in an increase in the pleural fluid LDH level of slightly more than 10% 48 hours after the biopsy (81).

When bloody pleural fluid is obtained, one might wonder whether the LDH measurement would be useful because RBCs contain large amounts of LDH. The presence of blood in the pleural fluid, however, usually does not adversely affect the measurement of the LDH. In one study, LDH isoenzyme analysis was performed on 12 pleural fluids that had contained more than 100,000 erythrocytes/mm^3. In only one effusion was the LDH-1 more than 5% above that in the serum, and the total pleural fluid LDH in that effusion was only 107 IU/L (40).

Although the total pleural fluid LDH level is not useful in distinguishing among various exudative pleural effusions, one might suppose that LDH isoenzymes would be useful in the differentiation. Three studies have shown that LDH isoenzymes have limited value in the differential diagnosis of exudative pleural effusions (40,142,143). All benign effusions with elevated pleural fluid LDH levels and most malignant effusions are characterized by a higher percentage of LDH-4 and LDH-5 in the pleural fluid than in the corresponding serum (40).

The increased amounts of LDH-4 and LDH-5 are thought to arise from the inflammatory WBCs in the pleural effusion (40). Approximately, one third of malignant pleural effusions have a different pleural fluid LDH isoenzyme pattern that is characterized by large amounts (>35%) of LDH-2 and less LDH-4 and LDH-5. None of the 31 benign exudates in one series had more than 35% LDH-2 (40). No relationship exists between the histologic type of the malignant pleural disease and the pleural fluid LDH isoenzyme pattern (40). At present, the only situation in which we obtain LDH isoenzyme analysis of pleural fluid is when there is a bloody pleural effusion in a patient who is clinically thought to have a transudative pleural effusion. If the LDH is in the exudative range and the protein is in the transudative range, the demonstration that most of the pleural LDH is LDH-1 indicates that the increase in the LDH is due to the blood.

pH AND P_{CO_2} MEASUREMENT

Measurement of the pleural fluid pH and P_{CO_2} is useful in the differential diagnosis of exudative pleural effusions. If the pleural fluid pH is less than 7.2, it means that the patient has 1 of 10 conditions: (a) complicated parapneumonic effusion, (b) esophageal rupture, (c) rheumatoid pleuritis, (d) tuberculous pleuritis, (e) malignant pleural disease, (f) hemothorax, (g) systemic acidosis, (h) paragonimiasis, (i) lupus pleuritis, or (j) urinothorax.

The pleural fluid pH is obviously influenced by the arterial pH. With transudative pleural effusions, the pleural fluid pH is usually higher than the simultaneous blood pH (43), presumably because of active transport of bicarbonate from the blood into the pleural space (44). If a low pleural fluid pH is discovered, the arterial pH should be checked to ensure that the patient does not have systemic acidosis. With certain exudative effusions, the pleural fluid pH falls substantially below that of the arterial pH. The explanation for the relative pleural fluid acidosis is as follows. The relationship between the pleural fluid pH and the blood pH depends on the extent to which the blood and pleural fluid P_{CO_2} and bicarbonate are in equilibrium. In conditions associated with pleural fluid acidosis, lactic acid accumulates in the pleural fluid (144,145), presumably from anaerobic glycolysis in the pleural fluid or tissues.

The hydrogen ions associated with the lactic acid combine with bicarbonate to form water and carbon dioxide. Accordingly, the pleural fluid P_{CO_2} increases and the pH decreases. Because the addition of 1 mEq of fixed acid to 1 L of pleural fluid results in an increase of 33 mm Hg in the P_{CO_2} but in a decrease of only 1 mEq in the bicarbonate concentration (146), pleural fluid acidosis is characterized by a P_{CO_2} that is increased proportionately more than the bicarbonate is reduced (43,144).

The increased pleural fluid P_{CO_2} could result from either an increased production of CO_2 or a decreased diffusion of CO_2 from the pleural fluid to blood or a combination of these factors. It is my belief that limited diffusion of CO_2 out of the pleural space is the predominant mechanism. In Figure 7.4A, it can be seen that changes in the arterial P_{CO_2} of a patient with a malignant pleural effusion and mild pleural fluid acidosis were not associated with changes in the pleural fluid P_{CO_2}.

Similarly, in a second patient, the administration of bicarbonate with an increase in the arterial pH from 7.40 to 7.59 did not change the pleural fluid pH or bicarbonate level (Fig. 7.4B). When pleural fluids are incubated at 37°C *in vitro*, no correlation exists between the rate of acid accumulation *in vitro* and the pleural fluid pH *in vivo* (146,147), with the possible exception of patients with complicated parapneumonic effusions in whom the rate of acid accumulation is high (147).

The pleural fluid pH is frequently not measured correctly (148–150). Chandler et al. (148) surveyed the methods by which pleural fluid pHs were measured at 277 acute care institutions in the southeastern part of the United States in 1998. They reported that the pleural fluid pH was measured with the blood gas machine in only 32% of the institutions, whereas it was measured with dip stick or pH indicator paper in 56% and by a pH meter in 12% (148). A survey of 267 pulmonologists in 2008 from the United States revealed that 39% of the physicians who use the pleural fluid pH in the management of parapneumonic effusions were wrong in their assumption that their laboratory used the blood gas machine to measure pleural fluid pH (150). It has been shown that neither pH indicator strip paper (148,151,152) nor pH meters (151) are sufficiently accurate for clinical use. Bowling et al. (153) recently reported similar results from North Carolina. In this study, only 2 of 11 hospitals measured pleural fluid pH with a blood gas analyzer. In a second study (153), 43% of 221 pulmonologists who use the pleural fluid pH were not aware that only pH's obtained with the blood gas machine are sufficiently accurate. The above studies demonstrate that it is important for physicians who order pleural fluid pH to know how their hospital measures the pleural fluid pH. The pH meter gives a reading that is approximately 0.20 to 0.30 too high because it measures the pH at

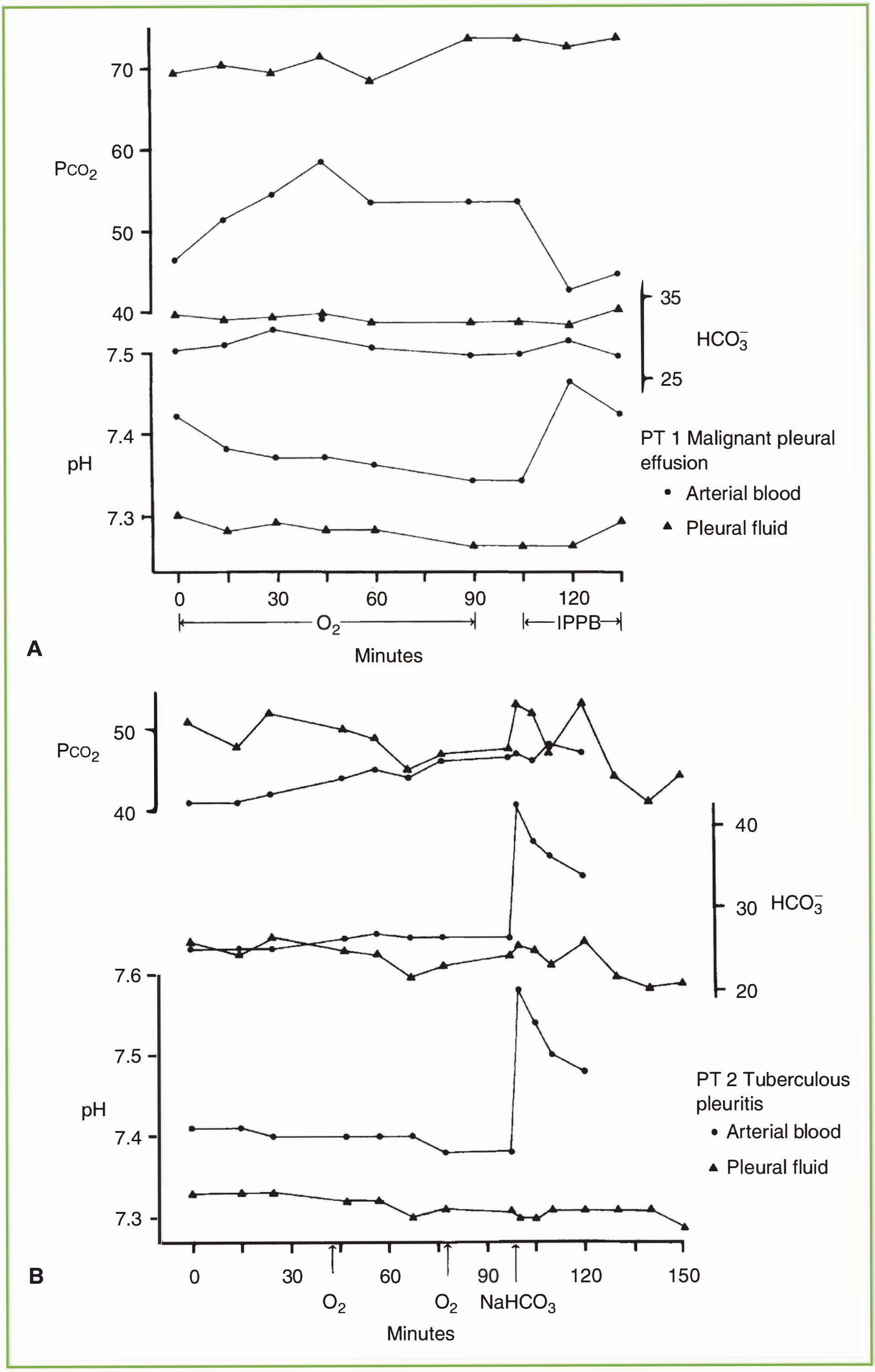

FIGURE 7.4 ■ Relationship between pleural fluid and arterial pH, P_{CO_2}, and HCO_3^-. **A**: Patient with a malignant pleural effusion in whom the administration of supplemental oxygen resulted in an increase in the Pa_{CO_2} from 46 to 58. Note that no concomitant change is seen in the pleural fluid P_{CO_2}. **B**: Patient with a slightly acidic pleural fluid. The administration of bicarbonate raised the arterial pH from 7.40 to 7.59 but had no influence on the pleural fluid pH. IPPB, intermittent positive pressure breathing.

room temperature (at which the Pco_2 is lower) and the sample comes in contact with room air, which allows the CO_2 to escape again lowering the Pco_2 (151). At times, laboratory personnel object to injecting the pleural fluid through blood gas machines for fear of the development of clots. This objection can be overcome if a clot-catching apparatus is inserted between the syringe and blood gas machine (151). The pleural fluid pH can also be measured accurately with a handheld analyzer such as the I-STAT Portable Clinical Analyzer that is used in intensive care units for measuring blood gases (149).

When the pleural fluid pH is used as a diagnostic test, it must be measured with the same care as arterial pH. The fluid should be collected anaerobically in a heparinized syringe (see Chapter 28). If the fluid is opened to room air, carbon dioxide will leave the fluid and the recorded pH will be falsely high (151,154). It appears that it is not necessary to put the fluid on ice if the pH is measured within an hour or so (154,155). Accurate results are also obtained when pleural fluid is transferred from a large syringe to a small heparinized syringe (156). If the effusion is small, injection of a local anesthetic into the pleural fluid may falsely lower the pH (157) as can residual lidocaine in the blood gas syringe (154). If frank pus is obtained at thoracentesis, one should not submit it for pH determination because the thick, purulent fluid may clog the blood gas machine and laboratory personnel may hesitate to analyze subsequent pleural fluids.

In general, pleural fluids with a low pH also have low glucose and high LDH levels (144). If the laboratory reports a low pH with normal glucose and low LDH levels, the pH measurement is probably in error. In a similar manner, a low glucose level with a normal pH and a low LDH is probably a laboratory error.

The only reason to measure the pleural fluid Pco_2 is to verify the pleural fluid pH because a low pleural fluid pH is almost always associated with a high Pco_2 (43,144). The pleural fluid Pco_2 adds no diagnostic value.

The pleural fluid pH is most useful in indicating the prognosis of patients with pneumonic effusions (see Chapter 9). If the pleural fluid pH is below 7.0, the patient invariably has a complicated parapneumonic effusion, and attempts should be made to remove all the pleural fluid with therapeutic thoracentesis or tube thoracostomy (118). If the pleural fluid pH is above 7.2, the prognosis of the patient is excellent and the pleural fluid need not be removed. The American College of Chest Physicians recommends using the pleural fluid pH to determine whether invasive procedures on the pleura are necessary for patients with parapneumonic effusions (158). If the patient has an infection with *Proteus* organisms, the pleural fluid pH may be elevated because these organisms produce ammonia by their urea-splitting ability, which can increase the pH (159). In patients with parapneumonic effusions, the pleural fluid pH may fall before the pleural fluid glucose level becomes depressed (117,160). It should be noted that the pH can vary markedly from locule to locule in patients with parapneumonic effusions (161).

The pleural fluid pH is also decreased with esophageal rupture (138,162). In fact, Dye and Laforet (162) concluded that a pleural fluid pH of less than 6.0 was highly suggestive of esophageal rupture. These workers attributed the low pleural fluid pH to the reflux of gastric acid through the rent in the esophagus into the pleural space. Subsequent studies in rabbits (163), however, have demonstrated that the pleural fluid pH becomes just as acidic after esophageal rupture if the esophagogastric junction is ligated. It appears that the low pleural fluid pH is due to infection in the pleural space rather than to acid reflux. Over the past few years, we have seen several patients with pleural infection without esophageal rupture in whom the pleural fluid pH was below 6.0.

In summary, esophageal rupture is associated with a low pleural fluid pH because of the concomitant pleural infection and not acid reflux. A pleural fluid pH below 6.0 is consistent with but not diagnostic of esophageal rupture.

Patients with pleural effusions secondary to both malignant disease and tuberculosis may have a low pleural fluid pH (43,144,164). When Light et al. wrote their first paper on pleural fluid pH (43), they concluded that the pleural fluid pH was useful in distinguishing tuberculous pleural effusions from malignant pleural effusions; a pleural fluid pH below 7.3 suggested tuberculosis, whereas a pleural fluid pH above 7.4 suggested malignant disease. Subsequent studies by others (145,164,165), however, and my own observations have not supported this conclusion. At present, I consider the pleural fluid pH valueless in distinguishing tuberculous pleural effusions from malignant pleural effusions. The pleural fluid pH does provide information about malignant pleural effusions because patients with a low pleural fluid pH have a shorter life expectancy and are less likely to have a favorable response to pleurodesis (121,165).

The pleural fluid pH is almost always less than 7.2 with rheumatoid pleural effusions (129). The pleural fluid pH with lupus pleuritis is usually above

7.35 (129), but occasionally, it is less than 7.2 (130). The pleural fluid pH tends to be low with both paragonimiasis (166) and Churg-Strauss syndrome (90), and these are the only two conditions in which a low pleural fluid pH is associated with pleural eosinophilia. Another situation in which the pleural fluid pH may be decreased is with a large hemothorax (43). The metabolism of the many RBCs in this condition, in conjunction with the atelectatic underlying lung, is the probable explanation for the decreased pH. Finally, the pleural fluid pH may be reduced with urinothorax (167). This is the only situation in which a transudative pleural fluid has a low pH without concomitant systemic acidosis.

TESTS FOR DIAGNOSING PLEURAL MALIGNANCY

Cytologic Examination of Pleural Fluid

Cytologic examination of pleural fluid is one of the most informative laboratory procedures in the diagnosis of pleural effusions because with it a definitive diagnosis can be made in more than 50% of patients with malignant disease involving the pleura. It is important to process pleural fluid specimens expeditiously when they are submitted for cytology. Specimens maintained at room temperature deteriorate markedly within 48 hours as do refrigerated specimens maintained for 96 hours (168). The optimal amount of pleural fluid to submit for direct smear and cell block preparations appears to be 150 ml although the diagnosis of malignancy can be established in most patients with 10 ml pleural fluid (169).

Malignant cells have several characteristics that differentiate them from other cells in the pleural fluid (61). Malignant cells in a given pleural effusion are recognizably similar to each other and are different from any nonmalignant cells in pleural fluid (Fig. 7.1C). Although the overall appearance of the malignant cells is similar, sometimes there is a marked variation in their sizes and shapes; one cell may have many times the diameter of its twin.

Frequently, malignant cells are large. The nuclei of malignant cells may exceed 50 μm in diameter, in contrast with mesothelial cell nuclei, which rarely exceed 20 μm in diameter. Small lymphocytes, by comparison, have a diameter of approximately 10 μm. The nucleoli of malignant cells are often large, exceeding 5 μm in diameter, whereas the nucleoli of nonmalignant cells in pleural fluid usually do not exceed 3 μm. Malignant cells have a high nucleocytoplasmic ratio. Indeed, the nuclear size of the cells in pleural effusions has been used diagnostically. Marchevsky et al. (170) performed a computer-assisted morphometric study of 48 pleural fluids including 20 benign fluids, eight mesotheliomas, and 20 carcinomas. If the mean nuclear diameter exceeded 10.5 μm or the mean nuclear diameter exceeded 9.3 μm, and the mean nuclear diameter divided by the cytoplasmic diameter exceeded 0.74, the patient had a malignant pleural effusion. This morphologic analysis was not able to separate carcinomas from mesotheliomas.

There are, however, cytologic characteristics that tend to be different for mesothelioma and adenocarcinomas. Stevens et al. (171) compared the cytologic characteristics of 44 cases of malignant mesothelioma and 46 cases of metastatic adenocarcinomas. They concluded that the following five features separate malignant mesothelioma from adenocarcinoma with more than 95% accuracy. Mesotheliomas tend to have true papillary aggregation, multinucleation with atypia, and cell-to-cell apposition, whereas adenocarcinomas tend to have acinus-like structures and balloon-like vacuolation (171).

Malignant cells sometimes aggregate, and large balls or clumps of cells are characteristic of adenocarcinoma (Fig. 7.1D). Although aggregates of 20 or more benign mesothelial cells occasionally occur, the bizarre, large, vacuolated cells with adenocarcinoma allow for a differentiation between these entities. Both malignant cells and macrophages may have vacuolation. Small numbers of mitotic figures frequently occur in benign effusions, and, accordingly, the presence of such figures is not indicative of malignant disease.

The accuracy of the cytologic diagnosis of malignant pleural effusions has been reported to be anywhere between 40% and 87% (172–174). Several factors influence the percentages in the various reports. First, in many patients with proven malignant disease and pleural effusion, the effusion is not related to malignant involvement of the pleura but is rather secondary to other factors such as CHF, pulmonary emboli, pneumonia, lymphatic blockade, or hypoproteinemia. In such patients, one cannot expect the result of the pleural fluid cytologic test to be positive. For example, it is unusual for the results of pleural fluid cytologic tests to be positive in patients with squamous cell carcinoma (41,61,175) because the pleural effusions are usually due to bronchial obstruction or lymphatic blockade. Second, the frequency of positive cytologic results depends on the tumor type. For example, with lymphoma, the cytologic examination was positive in

75% of patients with diffuse histiocytic lymphoma but in only 25% of patients with Hodgkin's disease in one series (176). The cytologic test is more frequently positive with adenocarcinomas than with sarcomas (175). Third, the accuracy of the results depends on the way in which the specimens are examined. If both cell blocks and smears are prepared and examined, the percentage of positive diagnoses will be greater than if only one method is used (177). It is recommended that the standard preparation of an effusion should include the preparation of a cellblock, and cytospins stained with Diff-Quik and Papanicolaou stains (178). Fourth, the more separate specimens submitted for cytologic examination, the higher the percentage of positive reports (41,176). In my own experience in patients with proven malignant disease involving the pleural space, the initial pleural fluid cytologic examination is positive in approximately 60% of patients, and if three separate specimens are submitted, nearly 80% of the patients will have positive results (42). The third specimen frequently contains fresher cells that allow the diagnosis to be made. Fifth, the incidence of positive diagnoses is obviously dependent on the skill of the cytologist. Sixth, the incidence of positive diagnoses is related to the tumor burden in the pleural space. A patient with a large tumor burden is more likely to have a positive pleural fluid cytology than is a patient with a small tumor burden.

Most tumors that are metastatic to the pleura have an epithelial origin whereas the cells that are normally in the pleural space (neutrophils, lymphocytes, mesothelial cells and macrophages) do not have an epithelial origin. Kielhorn et al. (179) reported that the yield with cytology was increased if epithelial cells were selected by immunomagnetic selection of cells binding to EpCAM antibodies. In their study of 59 effusions, the cytology alone was positive in 12 patients, but when the epithelial cells were identified with the EpCAM antibodies, the cytology became positive in 16 patients (179).

In summary, when three separate pleural fluid specimens from a patient with malignant pleural disease are submitted to an experienced cytologist, one should expect a positive diagnosis in approximately 80% of patients. Because it is important to prevent the pleural fluid specimen from clotting, about 0.5 mL heparin should be added to the syringe during a diagnostic thoracentesis (see Chapter 28). If a larger volume of pleural fluid is obtained during a therapeutic thoracentesis for submission to a cytologist, additional heparin should be added. From the examination of the exfoliated cancer cells, it is usually possible to classify the neoplasm accurately into its histologic type such as adenocarcinoma. Only occasionally is it possible to suggest with confidence the primary site of the neoplasm (175).

Nucleolar Organizer Regions

Nucleolar organizer regions (NOR) are loops of DNA in the nucleus that code for ribosomal RNA and are important in the synthesis of protein (180). These regions are associated with acidic nonhistone proteins that can be visualized by argyrophilic staining (AgNOR). In general, malignant cells have more AgNOR staining than benign cells (180). Although several papers have concluded that AgNOR staining is useful in separating benign and malignant effusions (181–184), there has been no standardization of the technique and there is overlap between benign and malignant cells. Until more research is conducted on AgNOR, it cannot be recommended.

Immunohistochemical Studies

With the development of the necessary technology for MAb, numerous papers have been published in the last 25 years that have assessed the diagnostic utility of MAb in the diagnosis of pleural malignancy.

The basis for this approach is the belief that there are antigens that are unique for benign mesothelial cells, adenocarcinoma cells, and malignant mesothelioma cells. If MAb are developed against these specific antigens, then positive identification of these cells can be made when tissue samples or cytologic preparations are incubated with the antibody and then counterstained with immunoalkaline phosphatase or using some similar method.

Many studies have compared the usefulness of the different antibodies in distinguishing the three different cells and these studies are summarized in a recent article by Ordonez (185). At the present time, the best markers for adenocarcinoma appear to be carcinoembryonic antigen (CEA), MOC-31, B72.3, Ber-EP4, BG8, and TTF-1, whereas the best markers for mesothelioma appear to be calretinin, keratin 5/6, podoplanin, and WT1 (185). TTF-1 has high specificity for lung carcinoma (185). It is important to realize that nonmalignant mesothelial cells will also stain positive for calretinin and cytokeratin 5/6 (186). Moreover, the sarcomatous type of mesothelioma is positive with calretinin less than 10% of the time (187). Ordonez concluded that D2-40 and podoplanin are the best immunohistochemical markers for epithelioid mesotheliomas (188).

When immunohistochemistry is used to differentiate adenocarcinoma from mesothelioma, a panel of MAb including two that stain with mesothelioma and two that stain with adenocarcinoma should be used (185). There are some additional points that need to be made about the immunohistochemical tests. For some of the antibodies, the pattern of the staining is very important. Therefore, it is necessary to have an experienced immunohistochemist performing the tests. There are new antibodies being evaluated continuously, and it is hoped that in the near future, antibodies will be developed that are specific for malignant mesothelioma and benign mesothelial cells. One preliminary report suggested that malignant mesotheliomas and benign mesothelial cells could be separated by the use of desmin and epithelial membrane antigen (EMA) (189). In this study 6/60 (10%) mesotheliomas and 34/40 (85%) of the benign mesothelial cells reacted to desmin, whereas 80% of mesotheliomas but only 20% of benign mesothelial cells reacted to EMA (189).

On occasions, one is faced with the situation where the cytology of the pleural fluid is positive but the location of the primary is unknown. Immunohistochemistry can suggest the location of the primary in some instances. For example, in one study of 67 patients with metastatic breast carcinoma, 21 of the patients (31.3%) stained positive with lactoferrin whereas no more than 15% of any other primaries were positive with this antibody (190). For 56 cases of ovarian carcinoma, immunohistochemistry was positive in 30% for CA-125 whereas no more than 12% of any of the other primaries were positive for this antibody (190). Probably the best antibody is thyroid transcription factor-1 (TTF-1). In one study, the pleural tissue stained positive for TTF-1 in 27/34 metastatic lung carcinomas but in none of the 48 other malignancies of the pleura (191). Cell-blocks from pleural effusions from patients with lung primaries also stain positive for TTF-1 (192). Small cell lung carcinoma also stains positive with TTF-1 as it does with chromographin A, a neuroendocrine marker (193). Nevertheless, there are no antibodies that are 100% specific for any primary (with the possible exception of TTF-1) and most antibodies are not very specific (190). In one case report of a patient with prostatic carcinoma metastatic to the pleura, the prostate specific antigen (PSA) was higher in the pleural fluid than it was in the serum (194).

It appears that immunohistochemistry is quite useful in establishing the diagnosis of a lymphomatous pleural effusion. Guzman et al. (195) performed immunocytochemical analysis with the peroxidase–antiperoxidase adhesive slide assay for detection of cell surface antigens using a broad panel of MAb in nine patients with pleural lymphoma. They were able to clearly recognize six cases of B-cell lymphoma, one case of Hodgkin's disease, and one case of hairy cell leukemia (195).

Immunohistochemical tests on cell blocks of pleural fluid or pleural biopsy specimens are available from Propath in Dallas, Texas, 1-800-258-1253 or www.propathlab.com.

Electron Microscopic Examination

The diagnosis of mesothelioma and metastatic carcinoma to the pleura is made in most instances by cytology and immunohistochemical assessment, but electron microscopy (EM) still plays a decisive role in some cases with unusual morphology or anomalous histochemical reactions. Accordingly, when malignant mesothelioma is suspected, a portion of the pleural biopsy specimen should be routinely fixed at the time of biopsy for possible subsequent processing for EM (196). EM has its greatest utility in differentiating metastatic adenocarcinoma from mesothelioma. The ultrastructural features of mesotheliomas are so characteristic as to be almost diagnostic. These characteristics include characteristic microvilli; the absence of microvillus core rootlets, glycocalyceal bodies, and secretory granules; the presence of intracellular desmosomes, junctional complexes, and intracytoplasmic lumina; and characteristic microvilli.

The appearance of the microvilli is the most important diagnostic feature. With adenocarcinoma, they are less abundant and are usually short and stubby, whereas with mesothelioma, they are numerous and are characteristically long and thin (197,198). These long microvilli can be visualized with light microscopy if the pleural fluid specimen is stained with the Ultrafast Papanicolaou stain (199).

Histochemical Studies

Immunohistochemical tests have replaced histochemical tests in distinguishing adenocarcinoma from mesothelioma in many laboratories. However, there are two primary histochemical tests that can be used to differentiate mesotheliomas from adenocarcinomas. The Alcian blue stain detects the acid mucins characteristic of mesothelioma (200). In one study, the Alcian blue stain was positive in 14 of 29 (47%) mesotheliomas (200), but in none of 44 patients with adenocarcinoma. The periodic acid-Schiff stain after diastase digestion (PAS-D) detects neutral mucins, which are diagnostic of adenocarcinomas. In one study the PAS-D stain was positive in 27 of 44 (61%) patients with adenocarcinoma but in no patients with mesothelioma (200). In summary, if the cells stain positive with PAS-D, the patient in all probability has an adenocarcinoma. If the cells stain positive with Alcian blue, the patient in all probability has a mesothelioma. If the cells stain positive with neither, no conclusion can be made.

Tumor Markers in Pleural Fluid

The possibility of establishing the diagnosis of pleural malignancy by demonstrating an elevated level of a tumor marker in the pleural fluid has been the subject of many publications. The tumor markers that have been evaluated have included (201–212), carbohydrate antigens CA 15-3 (201,205–209,212–214), CA 19-9 (201,205,207,208,215), CA 549 (212), CA 72-4 (201,205,212), cytokeratin 19 fragments (CYFRA 21-1) (201,202,206,207,216), cancer antigen 125 (CA 125) (206,207), sialyl stage-specific antigen (217,218), neuron-specific enolase (201,203,219), squamous cell carcinoma antigen (201,203), HER-2/neu (220) and telomerase (221,222).

In general, I do not recommend that tumor markers be used in the evaluation of patients with undiagnosed pleural effusions (223). Although an elevated level of any of the tumor markers is very suggestive of malignancy, specificity is not high enough to establish the diagnosis. Although there is no doubt that the median levels of the different tumor markers are significantly higher in patients with malignancy than in patients with benign pleural effusions, there is always some overlap in the values. If the cutoff level for a tumor marker is set high enough that the level is exceeded by none of the benign effusions, then the test tends to be very insensitive. The same criticism could be made of tests on the pleural fluid for tuberculosis, such as adenosine deaminase (ADA) or interferon-gamma. Nevertheless, I rely heavily on these tests to establish the diagnosis of tuberculosis. The primary difference in the two situations is that it would be disastrous to wrongly make the diagnosis of malignancy with a tumor marker because the patient is essentially told that he or she has only 90 days to live. If the diagnosis of tuberculosis is wrongly established, the patient is not sentenced to death but rather to taking antituberculous medications for 6 to 9 months.

The best study to date was reported by Porcel et al. (206). They measured the pleural fluid levels of CEA, CA-125, CA 15-3 and cytokeratin 19 fragments in 416 patients including 166 with definite malignant effusion, 77 with probable malignant effusions, and 173 with benign effusions. Cutoff levels were established such that the levels in all 173 benign effusions were below the cutoff level. Using these cutoff levels, 54% of the malignant effusions were classified as malignant and more than one third of the cytology-negative malignant pleural effusions could be identified by at least one marker (206). These authors suggested that the main use of measuring tumor markers in the pleural fluid would be to select patients (those with the higher levels) for more invasive studies (206).

Soluble Mesothelin Related Protein (SMRP)

It has been suggested that elevated pleural fluid levels of SMRP are useful in the diagnosis of malignant mesothelioma (224–226). Creaney et al. (224) measured the SMRP levels in pleural fluids from 192 patients presenting to a respiratory clinic including 52 with malignant mesothelioma, 56 with nonmesotheliomatous malignancies and 84 benign effusions. The pleural effusions from the patients with mesothelioma had significantly higher concentrations of SMRP than did the other patients (224). However, the SMRP in patients with sarcomatoid mesothelioma did not differ significantly from nonmalignant effusions (224). Davies et al. (225) measured pleural fluid SMRP in 24 patients with mesothelioma, 67 patients with pleural metastases and 75 patients with benign conditions. Using ROC curve analysis, pleural fluid SMRP had an AUC of 0.878 in its ability to differentiate between patients with mesothelioma and all other diagnoses at an optimal cutoff value of 20 nM (225). At this cutoff, the diagnostic sensitive and specificity were 0.71 and 0.90, respectively (225). Again, the levels of SMRP were lower in patients who had sarcomatoid mesothelioma (225). Adenocarcinomas accounted for 12 of the 13 false positives. The above two studies demonstrate that pleural fluid SMRP measurement provide additional information to cytology. However, tissue confirmation of mesothelioma is indicated in most situations.

Oncogenes

The development of cancer is a multistep process in which multiple genetic alterations must occur. The transforming genes are collectively called *oncogenes*. The oncogenes may be related to viruses, environmental carcinogens, or spontaneous mutations. Because the oncogenes are associated with the development of malignancy, one might hypothesize that patients with pleural malignancy would have cells in their pleural fluid containing oncogenes.

There have now been several studies testing this hypothesis. Zoppi et al. (227) attempted to detect the p53 protein in 34 embedded blocks of neoplastic fluids and 30 nonneoplastic effusions. They reported that 11 (34%) of the tumor fluids were positive, whereas all the benign fluids were negative (227). Mayall et al. (228) reported similar findings for p53. Tawfik and Coleman (229) reported that benign and malignant effusions did not differ significantly in their expression of the *CMYC* oncogene. Athanassiadou et al. (230) reported that although 21 of 24 malignant effusions (87%) were positive for the *CHARAS* oncogene, the diagnostic usefulness of the test was limited because 6 of 16 benign effusions (37%) also tested positive.

Hyaluronic Acid

Pleural fluid from patients with mesotheliomas is sometimes abnormally viscid. The increased viscosity in such fluids is due to the presence of increased amounts of hyaluronate, which was previously called *hyaluronic acid*. Rasmussen and Faber (231) examined the diagnostic usefulness of pleural fluid hyaluronic acid levels in 202 exudates including 19 malignant mesotheliomas. These investigators found that 7 of 19 pleural fluids (37%) from patients with malignant mesotheliomas had hyaluronate concentrations above 1 mg/mL, whereas none of the other pleural fluids had hyaluronate levels above 0.8 mg/mL. Nurminen et al. (232) assayed the levels of hyaluronate in 1,039 pleural effusions including 50 from patients with mesothelioma. They reported that when a cutoff level of 75 mg/mL was used, the assay specificity for malignant mesothelioma was 100% and the sensitivity was 56%. Two more recent papers (233,234) have reported that the mean levels of hyaluronate are comparable in patients with mesothelioma and metastatic adenocarcinoma. The explanation for the discrepant results appears to be methodologic (235). The measurements of Nurminen were obtained with high-pressure liquid chromatography (HPLC), whereas those of the latter two groups were by radioimmunoassay. Unfortunately, the only commercially available measurements in the United States of which I am aware use radioimmunoassay (Specialty Laboratories, San Diego, CA, USA, and SmithKline Beecham Clinical Laboratories, Philadelphia, PA, USA). Until the results with this assay are verified, it cannot be recommended on a routine basis.

Lectin Binding

Lectins are a class of glycoproteins of nonimmune origin that bind specifically to carbohydrate groups found ubiquitously in various biologic products. Kawai et al. (236) investigated lectin binding in 23 pleural mesotheliomas, 6 effusions with reactive mesothelial cells, and 28 well-differentiated pulmonary adenocarcinomas. In this study, some of the lectins were much more likely to bind to adenocarcinomas than to reactive mesothelial cells or mesothelioma cells. These workers could not find significant differences in lectin binding between mesotheliomas and reactive mesothelial cells. Additional research in this area may well demonstrate that studies of lectin binding are useful diagnostically. At the present time, such studies should be considered experimental.

Flow Cytometry

Flow cytometry provides a method for the rapid quantitative measurement of nuclear DNA. It has been proposed as a suitable tool for differentiating between benign and malignant cells because most malignant tumor cells possess an abnormal number of chromosomes (aneuploidy), and consequently an abnormal DNA content (DNA aneuploidy) (237). However, a substantial percentage of metastatic adenocarcinomas and most malignant mesotheliomas are diploid, as demonstrated through flow cytometry (180,238–240). Moreover, some benign effusions are aneuploid (239). Accordingly, the routine use of flow cytometry to quantitate nuclear DNA levels for the differentiation of benign and malignant effusions cannot be recommended.

Flow cytometry can also be used to identify the surface markers of lymphocytes rapidly and specifically using immunocytometry (241). Accordingly, the cell lineage (T or B cells) and the clonality of a population of lymphocytes can be determined. These techniques can therefore be used to establish the diagnosis of pleural lymphomas and are recommended in lymphocytic pleural effusions on which the diagnosis of lymphoma is a consideration (241).

Flow cytometry can also be used with the cells labeled with a panel of antibodies such as MOC-31, EMA, CEA, B72.3, keratin, desmin, CA-125, IMP3 and GLUT-1 (242,243). These studies appear to be complementary to studies using only cytology (242,243).

Chromosomal Analysis

Abnormalities undoubtedly exist in the chromosome number and structure in some patients with malignant pleural effusions (244,245). Malignant cells have more chromosomes and marker chromosomes, which are chromosomes with structural abnormalities (translocation, deletion, acentric, dicentric, inversion, isochromosome, or ring) (244). It remains to be demonstrated that there is a place for chromosomal analysis in the routine examination of pleural fluid. However, there may be a place for fluorescence in situ hybridization (FISH) to demonstrate chromosomal abnormalities.

In the last few years, a new methodology called *FISH* has been introduced (246). With this technique prespecified chromosomal aberrations can be visualized in cytologic smears (246). Fiegl et al. (247) used cytology and FISH techniques on 194 effusions from patients with malignancy. They reported that cytology was positive in 99 (51%) whereas FISH was positive in 116 (60%) (247). One of the two tests was positive in 133 (69%) of the effusions (247). They concluded that the FISH technique was complimentary to cytology in establishing the diagnosis of malignant pleural effusion (247). The FISH technique on the pleural fluid is also useful in distinguishing mesothelioma from benign effusions (248).

Proteomics

Proteomics involves identifying all the proteins present in a given specimen and then identifying those proteins that are unique to a certain condition. There have been a couple of preliminary articles exploring the feasibility of using proteomics for the diagnosis of malignant pleural effusions (249,250). Unique proteins have been identified but it is still too early to determine whether proteomics will turn out to be diagnostically useful. Another preliminary study (251) used proteomics to analyze the proteins from exosomes isolated from pleural effusions and reported that they were able to identify some proteins that were not previously reported.

TESTS FOR DIAGNOSING PLEURAL TUBERCULOSIS

Adenosine Deaminase Measurement (ADA)

Measurement of the ADA level in pleural fluid is diagnostically useful because ADA levels tend to be higher in tuberculous pleural effusions than in other exudates (252–256). ADA is the enzyme that catalyzes the conversion of adenosine to inosine. In general, a cutoff level of between 40 and 45 U/L is used with levels above this being indicative of tuberculosis. The higher the level, the more likely the patient is to have tuberculosis. Liang et al. (255) performed a meta-analysis of 63 articles evaluating the diagnostic usefulness of ADA that included 2,796 patients with tuberculous pleuritis and 5,297 patients with other diseases. They reported that the mean sensitivity was 0.92, the mean specificity was 0.90, the mean positive likelihood ratio was 0.903 and the mean negative likelihood ratio was 0.10 (255). In the largest series from a single institution, Porcel et al. (256) reported measurements of the pleural fluid ADA. In their study of 2,104 pleural effusions including 221 with tuberculous pleuritis, the sensitivity was 0.93, the specificity was 0.90, the positive likelihood ratio was 10.05 and the negative likelihood ratio was 0.07 (245). Valdés et al. (254) reported their results on pleural fluids from 405 patients, including 91 due to tuberculosis, 110 due to malignancy, 58 due to pneumonia, 10 due to empyema, 88 transudates, and 48 miscellaneous. Their results were very similar to those of previous workers with the exception that empyemas also had very high ADA levels (Fig. 7.5). Measurement of the ratio of the pleural fluid to the serum ADA is much less useful diagnostically (254).

The pleural fluid ADA is also elevated in patients with tuberculous pleuritis who are immunosuppressed. The levels of ADA in patients with and without AIDS are comparable (257) and renal transplant patients who develop a tuberculous pleural effusion have an elevated pleural fluid ADA level (258).

Some caution must be used in relying on ADA levels exclusively to establish the diagnosis of tuberculous pleuritis. The two main diseases that cause an elevated ADA in addition to tuberculosis are rheumatoid pleuritis and empyema (254,259). If the diagnostic criteria for tuberculous pleuritis also include a pleural fluid lymphocyte-to-neutrophil ratio greater than 0.75, the specificity of the test is increased (260). High pleural fluid ADA levels have also been reported with a very small percent of other neoplasms (261), with Q fever (262), with brucellosis (263) and with Legionnaire's disease (264).

The pleural fluid ADA level can be used to exclude the diagnosis of tuberculous pleural effusions in patients with undiagnosed lymphocytic pleural effusions. Two series (265,266) measured the pleural fluid ADA levels in 506 patients with lymphocytic pleural effusions not due to tuberculosis. Only 10 patients (2%) had an ADA level above 40 U/L. The diagnoses in these 10 patients included 3 cases of lymphoma, 3 cases of parapneumonic effusions, 2 bronchogenic carcinomas, 1 mesothelioma, and 1 idiopathic pleural effusion (265,266). The pleural effusions that occur

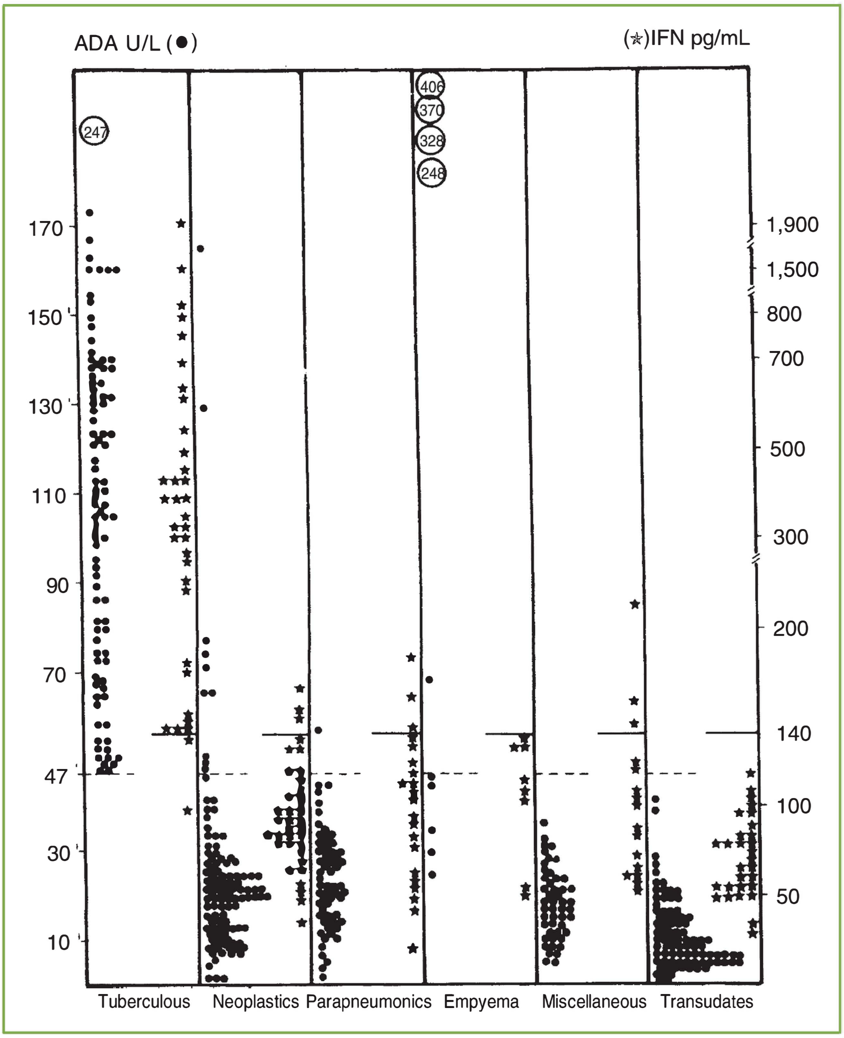

FIGURE 7.5 ■ Pleural fluid adenosine deaminase (ADA) and interferon (IFN)-gamma levels in 430 cases. *(From Valdés L, San Jose E, Alvarez D, et al. Diagnosis of tuberculous pleurisy using the biologic parameters adenosine deaminase, lysozyme, and interferon-gamma.* Chest. *1993;103:458–465, with permission.)*

after CABG surgery are typically lymphocytic, but their ADA levels are below 40 U/L (265).

Several papers have been written on the diagnostic utility of ADA isoenzymes. ADA has two isoenzymes, ADA1 and ADA2 (267–270). ADA1 is ubiquitous and is produced by lymphocytes, neutrophils, monocytes, and macrophages (268). In contrast, ADA2 exists only in monocytes and macrophages. The increase in ADA activity with tuberculous pleuritis is mainly due to ADA2, which is surprising because most of the cells in the pleural fluid are lymphocytes. These observations suggest that the origin of the pleural fluid ADA is probably in the pleural tissues rather than the cells in the pleural fluid. There is one report in which the use of an ADA1-to-ADA total ratio of less than 0.42 increased the accuracy with which the diagnosis of tuberculous pleuritis was established (269). In most cases, ADA isoenzymes are not needed to establish the diagnosis of tuberculosis. However, in certain instances they can be quite useful.

An added advantage of ADA in the diagnosis of pleural effusions is that if 0.9 mL pleural fluid is added to a test tube containing 0.10 mL of a mixture of 50% glycerol and 50% ethylene glycol, the pleural fluid can be mailed by regular mail with no loss of ADA activity (271). If the pleural fluid is stored at

4 degrees C or -20 degrees C, the level of ADA remains stable up to 28 days (272).

In view of the foregoing information, it is recommended that facilities in which a sizable percentage of the cases of pleural effusions are due to tuberculosis develop the faculty to perform ADA assays on pleural fluid. An ADA level above 70 U/L in a patient who does not have an empyema or rheumatoid arthritis (RA) is essentially diagnostic of tuberculous pleuritis. An ADA level above 40 is suggestive of tuberculosis, and the higher the ADA, the more likely the diagnosis of tuberculous pleuritis.

Interferon-Gamma

Measurement of the pleural fluid levels of interferon-gamma is useful in the diagnosis of tuberculous pleuritis because patients with tuberculous pleuritis tend to have higher levels of interferon-gamma than do other types of exudates (273–278). Jiang et al. (278) performed a meta-analysis on the diagnostic usefulness of pleural fluid interferon-gamma levels in 22 studies with 782 patient with tuberculous pleuritis and 1,319 patients with pleural effusions due to other diseases. They reported that the mean sensitivity was 0.89, the mean specificity was 0.97, the mean positive likelihood ratio was 23.45 and the mean negative likelihood ratio was 0.11 (278).

Individual studies have reported similar results. Pleural fluid interferon-gamma levels from the 145 patients reported by Valdés et al. are shown in Figure 7.5. As can be seen from this figure, 26 of 35 patients (74%) with tuberculous pleurisy had interferon-gamma levels above 200 pg/mL, whereas only 1 out of 110 other effusions that were not empyemas had an interferon-gamma level that exceeded this. Villena et al. (277) reported comparable results in the pleural fluids of 595 patients, including 82 with tuberculous effusions. She reported that an interferon-gamma level of 3.7 IU/mL (as measured by radioimmunoassay) had a sensitivity of 0.98 and a specificity of 0.98 (277) for the diagnosis of tuberculous pleuritis.

Interferon-gamma is produced by the $CD4^+$ lymphocytes from patients with tuberculous pleuritis (275). The production of interferon-gamma appears to be a useful defense mechanism. Interferon-gamma enhances polymyristate acetate–induced hydrogen peroxide production in macrophages, facilitating elimination of intracellular parasites. This lymphokine also inhibits mycobacterial growth in human monocytes (275).

In view of the information mentioned in the preceding text, it appears that measurement of the interferon-gamma level is very useful in the diagnosis of tuberculous pleuritis. One must be careful in interpreting the results from a given laboratory, however, as different laboratories report their results in different units. In the United States, pleural fluid interferon-gamma levels can be obtained from Immunoscience Laboratory in Los Angeles, 310-667-1077, www.immuno-sci-lab.com.

Which test should be used to establish the diagnosis of tuberculous pleuritis? Greco et al. (279) reviewed all English language studies from 1978 to November 2000. The studies included 4,738 patients on whom ADA was measured and 1,189 patients on whom interferon-gamma was measured. These researchers reported that the maximum joint sensitivity and specificity for ADA was 93%, whereas it was 96% for interferon-gamma (279). Because there is not much difference in the performance of the two tests and as ADA is much less expensive, ADA appears to be the preferred test.

Interferon-gamma release assays (IGRA)

IGRA on the blood were designed to diagnose latent tuberculosis (280). There are two commercially available assays the QuantiFERON TB Gold (Qtf Gold; Cellestis Ltd., Carnegie, Victoria, Australia) and the T SPOT.TB (Oxford Immunotec Ltd., Abington, UK). These tests incubate whole-blood or isolated peripheral blood mononuclear cells with Mycobacterium tuberculosis specific antigens and assay the amount of interferon-gamma released.

The IGRA are inferior to the pleural fluid interferon-gamma levels (280–281). Zhou et al. (281) performed a meta-analysis of the available reports on IGRA on the pleural fluid for diagnosing tuberculous pleuritis. They analyzed seven reports with a total of 213 patients with tuberculous pleuritis and 153 patients with pleural effusions of other etiologies and reported that the mean sensitivity was 0.75, the mean specificity was 0.82, the mean positive likelihood ratio was 3.49 and the mean negative predictive ratio was 0.24 (281). Since these results are markedly inferior to those with ADA or interferon-gamma, the IGRA should not be used in assessing whether patients have tuberculous pleuritis.

Polymerase Chain Reaction

The role of the polymerase chain reaction (PCR) and other nucleic acid amplification (NAA)–based tests in the diagnosis of tuberculous pleuritis remains to be established. There are several NAA tests available for the rapid detection of *Mycobacterium tuberculosis* in clinical specimens. At present, the U.S. Food and Drug Administration (FDA) has not given its approval for the use of these tests on extrapulmonary materials (282).

Pai et al. (283) performed a meta-analysis on 14 studies utilizing NAA-based tests for the diagnosis of tuberculous pleuritis using three different commercial kits including 127 patients with tuberculous pleuritis and 1,400 patients with pleural effusions due to other diseases. They reported that the mean sensitivity was only 0.62 while the mean specificity was 0.98 (283). The mean positive likelihood ratio was 25.4 while the mean negative likelihood ratio was 0.40 (283).

In general, PCR on pleural fluid has been less sensitive than the PCR on other specimens, possibly due to the low numbers of tuberculous bacilli present in pleural fluid and possibly due to the manner in which the DNA is extracted (284). PCR has also been tried on pleural biopsy specimens and the results have not been particularly promising (285).

In view of their low sensitivity, the use of the PCR or other NAA-based tests should be considered investigative. Certainly, the expense associated with these tests is much greater than that associated with pleural fluid ADA measurements. In the future, it is possible that these tests will be widely used in the diagnosis of tuberculous pleuritis.

C-Reactive Protein

Patients with tuberculous pleuritis tend to have higher pleural fluid levels of C-reactive protein (CRP) than do patients with other lymphocytic pleural effusions. Garcia-Pachon et al. (286) measured the CRP in 144 patients with lymphocytic pleural effusions including 20 with tuberculosis, 69 with malignancy, 38 with transudates, and 17 with other benign exudates. They found that CRP levels greater than 50 mg/L had a high specificity for tuberculosis (95%) and levels less than 30 mg/L had a high sensitivity (95%) for excluding tuberculosis. In a second study (287) of 148 patients with lymphocytic exudative pleural effusions including 55 with tuberculous pleuritis, a pleural fluid CRP level above 30 mg/L had a sensitivity of 72% with 93% specificity. The advantage of the CRP test is that it is inexpensive and widely available. However, it does not appear to be as accurate as ADA levels in making the diagnosis of tuberculous pleuritis. Moreover, the mean pleural fluid levels in patient with parapneumonic effusions is greater than that in tuberculous pleural effusions (288).

Lysozyme

The level of lysozyme in the pleural fluid tends to be higher in the pleural fluid from patients with tuberculous pleuritis than in other types of exudates (242,243,289,290). Lysozyme is a bacteriolytic protein with a low molecular weight distributed extensively in organic fluids. In general, the lysozyme levels in tuberculous pleural effusions are greater than those in malignant pleural effusions, but there is so much overlap that the pleural fluid levels themselves are not particularly useful diagnostically. There was one report that suggested that the ratio of the pleural fluid to the serum lysozyme level was useful in separating the two diseases (290). Verea Hernando et al. (290) reported the results for 54 patients with tuberculous pleural effusions and 35 patients with malignant pleural effusions. All of the patients with tuberculosis had a ratio above 1.2, whereas only 1 of the 35 patients (3%) with malignant pleural effusions had ratios above 1.2. A recent study by Valdés et al. (254) did not reproduce these good results; nearly one third of the patients with tuberculous pleuritis had a lysozyme ratio less than 1.1, and many other exudates had ratios that exceeded this value. In this latter article, both the interferon-gamma and the ADA were superior to the lysozyme ratio in differentiating tuberculous exudates from nontuberculous exudates (254). At the present time, the routine measurement of pleural fluid lysozyme to help establish the diagnosis of pleural tuberculosis is not recommended.

Procalcitonin

Serum procalcitonin levels are a useful marker of severe systemic bacterial infection (291). Serum levels of procalcitonin are useful in the early diagnosis of systemic bacterial infection and sepsis, assessment of the severity and prognosis of sepsis and multiple organ failure, and the differentiating of bacterial and viral infections (291). For procalcitonin levels in the pleural fluid, it has been shown that the highest mean levels are with empyema followed by parapneumonic effusions and then tuberculous pleurisy and malignant pleural effusion (291,292). However, there is so much overlap that the diagnostic value of procalcitonin is very limited.

Tuberculous Antigens and Their Antibodies

The possibility of establishing the diagnosis of tuberculous pleuritis by the demonstration of tuberculous antigens or specific antibodies against tuberculous proteins in the pleural fluid has also been investigated.

Two reports have evaluated the diagnostic utility of measuring the levels of different tuberculous antigens in the pleural fluid (292,293). Although the mean levels of tuberculous antigens were higher in the pleural fluid of patients with tuberculous pleuritis than in the pleural fluid of other patients, there was so much overlap that the test was of little diagnostic use. In a similar vein, there have been at least eight separate reports (294–301) that have indicated that patients with tuberculous pleural effusions tend to have higher levels of specific antituberculous antibodies in their pleural fluid than do patients with other types of exudative effusions. The source of the antibodies, however, is apparently the serum rather than local antibody production in the pleural space in most instances (296), although in one study, the pleural fluid levels of IgM antibodies against mycobacterial antigen A60 were higher in the pleural fluid than in the serum and were higher in patients with tuberculous pleuritis than in those without tuberculous pleuritis (301). If these results can be confirmed, this test may prove useful in identifying patients with tuberculous pleuritis.

IMMUNOLOGIC STUDIES

Because nearly 5% of patients with RA (302) and 50% of patients with systemic lupus erythematosus (SLE) have pleural effusions sometime during the course of their disease, and because such effusions may be present before the underlying disease is obvious (129,302), it is important to consider these diagnostic possibilities in patients with exudative pleural effusions of undetermined origin. Numerous papers have assessed the diagnostic utility of various immunologic measurements of the pleural fluid in establishing these diagnoses.

Rheumatoid Factor

Berger and Seckler (303) first reported that rheumatoid factor (RF) was elevated in the pleural fluid of patients with rheumatoid pleuritis. Subsequently, Levine et al. (304) studied pleural fluid RF levels in 65 patients with pleural effusions and found that 41% of patients with bacterial pneumonia and 20% of patients with carcinoma had pleural fluid RF titers equal to or greater than 1:160. In seven of the 65 fluids, the pleural fluid RF titers were greater than the serum titers, but the pleural fluid titer was greater than 1:640 in only one of the patients. These workers concluded that the RF titers in pleural fluid were not useful diagnostically. Halla et al. (129), however, found that the pleural fluid RF was elevated in 11 of 11 seropositive patients with rheumatoid pleural effusions. In each patient, the pleural fluid RF titer was equal to or greater than 1:320 and was equal to or greater than that in the serum. In view of the last-mentioned study, I recommend that RF titers be determined in pleural fluid when the diagnosis of rheumatoid pleuritis is considered. The demonstration of a pleural fluid RF titer equal to or greater than 1:320 and equal to or greater than the serum titer is strong evidence that the patient has a rheumatoid pleural effusion.

Antinuclear Antibodies

In the earlier editions of this book, I stated that measurement of the antinuclear antibody (ANA) levels in pleural fluid appeared to be the best test for establishing the diagnosis of lupus pleuritis. Two subsequent studies (305,306) have cast doubt on this statement. Khare et al. (305) measured pleural fluid ANA levels using the Hep-2 cell line on 82 pleural fluids including 8 with SLE. Six of the eight patients (75%) with SLE had high (>1:320) titers of ANA in their pleural fluid with a homogeneous staining pattern. In none did the pleural fluid ANA titer differ by more than one dilution from the serum titer. The other two patients with SLE had alternative explanations for their pleural effusions. However, 8 of the remaining 74 patients (10.8%) had positive pleural fluid ANA titer (>1:40) and 2 had a homogeneous pattern. In the patients with SLE, pleural fluid analysis of anti-ssDNA, anti-dsDNA, anti-Sm, anti-SSA, and anti-SSB antibodies reflected the findings in the serum (305).

A more recent study evaluated the ANA titers in 126 pleural fluids using the HEP-2 cell line as substrate (306). Again all of the pleural fluids from the patients with SLE had a high ANA titer (>1:160), but in this study, the pattern could be either homogeneous or speckled (306). Moreover, pleural fluid ANA titers were greater than 1:160 in 13 other patients, 11 of whom had malignant pleural effusions. The pleural fluid and the serum ANA titers were closely correlated in all patients. The explanation for the discrepancy between these two recent studies and two older studies (130,307), which suggested that the pleural fluid ANA levels were very useful, is not known but it may be related to the cell line used for the assays.

Nevertheless, the two more recent studies suggest that measurement of the pleural fluid ANA titer adds essentially nothing to measuring the serum ANA titer in the differential diagnosis of pleural effusions.

Lupus Erythematosus Cells

In the past, the demonstration of LE cells in a pleural effusion was thought to be diagnostic of lupus pleuritis (130). LE cells are formed when polymorphonuclear leukocytes ingest extracellular nuclear material to form a polymorphonuclear leukocyte with a large inclusion of nuclear material. Most pleural effusions secondary to SLE contain LE cells (307). Wang et al. (308) reported that LE cells were present in the pleural fluid in 8 of 10 patients (80%) with SLE and polyserositis but in none of 112 pleural fluids with other etiologies (308). Noteworthy was the observation that the LE cell test results were identical in the serum and the pleural fluid. This suggests that the diagnosis of SLE can be established by an LE test on the serum, but an LE test on the pleural fluid adds nothing diagnostically. Moreover, with the development of better immunologic tests for SLE, LE preparations are being performed less and less frequently and laboratories therefore become less proficient in performing this test. Testing for LE cells is not recommended unless the laboratory is proficient in this test. It should also be noted that false-positive LE cell results have been reported (309).

Rheumatoid Arthritis Cells

The cytologic picture with rheumatoid pleuritis can help establish the diagnosis of rheumatoid pleuritis. Nosanchuk and Naylor (310) initially described a unique cytologic picture characterized by large elongated cells, giant round or oval nucleated cells, and a background of amorphous granular material. These elongated cells are sometime called *comet tadpole* or *comet-shaped cells* (Fig. 7.6). This cytologic picture is thought to be highly specific and pathognomonic of rheumatoid pleuritis (311).

Complement Levels

Most patients with pleural effusions secondary to either SLE or RA have reduced pleural fluid complement (Fig. 7.6), irrespective of whether whole complement (CH50) (312,313), C3 (129), or C4 (129,312) is measured. The measurement of pleural fluid complement does not absolutely separate patients with SLE or RA from patients with other exudative pleural effusions, as shown in Figure 7.7,

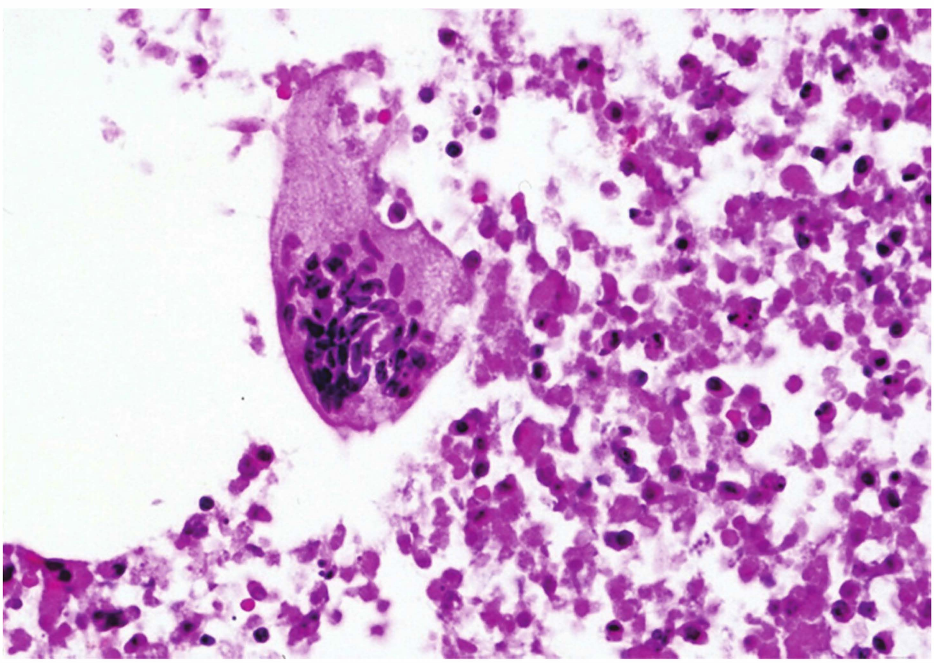

FIGURE 7.6 ■ Multinucleated tear drop–shaped cell from a patient with rheumatoid pleuritis, which is thought to be pathognomonic of this disease.

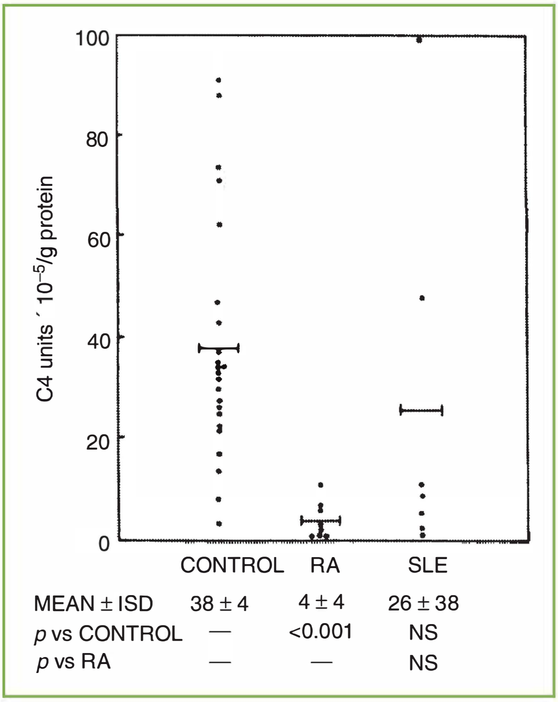

FIGURE 7.7 ■ Levels of hemolytic C4 adjusted for total protein in control, rheumatoid arthritis (RA), and systemic lupus erythematosus (SLE) pleural fluids. Note that the pleural fluid from the patients with RA and SLE has lower levels of hemolytic C4. NS, not significant. *(From Halla JT, Schrohenloher RE, Volanakis JE. Immune complexes and other laboratory features of pleural effusions.* Ann Intern Med. *1980;92:748–752, with permission.)*

regardless of whether CH50, C3, or C4 is measured. Nevertheless, a CH50 level below 10 U/mL (312) or a C4 level below 10 times 10^{-5} U/g protein (129) is seen with most patients with either RA or SLE and rarely with other diseases. Because the serum ANA levels and the RF titers appear to be more specific and more sensitive at identifying individuals with pleural effusions due to RA or SLE, measurement of pleural fluid complement levels is not routinely recommended.

There have been several articles that have evaluated the diagnostic utility of complement activation products in pleural fluid. Two studies have demonstrated that the pleural fluid levels of SC5b-9, which is a product of C3 activation, are elevated in tuberculous pleural fluid as opposed to malignant pleural effusions or transudates (314,315) and that the SC5b-9 levels are highest in patients with rheumatoid disease (315). Another study (316) demonstrated that the SC5b-9 levels tended to be higher in complicated than uncomplicated parapneumonic effusions. However, the overlap between the various groups limits the diagnostic usefulness of this test.

Immune Complexes

Patients with pleural effusions secondary to RA or SLE also have higher pleural fluid levels of immune complexes than patients with effusions due to other causes (129,317,318). The differentiation of pleural effusions secondary to SLE and RA from other exudative pleural effusions is less distinct when pleural fluid immune complex levels are used than when pleural fluid complement is used because a substantial percentage of other exudative pleural effusions have elevated pleural fluid immune complex levels (129,317). The presence of pleural fluid immune complexes also depends on the assay system (129). Patients with SLE have higher levels of immune complexes when the assay is performed with the Clq component of complement or the Raji cell assay than with monoclonal RF. With RA, the level of immune complexes in the pleural fluid is higher than that in the simultaneously obtained serum, whereas in other disease states, the serum level of immune complexes is higher (129). Because measurement of the immune complex level appears to add no information to that obtained from measuring complement levels, it is recommended that pleural fluid immune complex determinations be performed only in research situations.

LIPID STUDIES

Pleural fluid is occasionally milky or opalescent. This opalescence is sometimes mistakenly attributed to myriad WBCs in the pleural fluid, and the patient is treated for an empyema. This mistake is not made if the supernatant of the centrifuged pleural fluid is examined. In empyema, the supernatant is clear, whereas in a chylous or chyliform effusion, the supernatant remains cloudy or milky. Some pleural fluids with high lipid content also contain numerous RBCs, and their color is accordingly red or brown. The supernatants of all pleural fluids should be examined for turbidity. Lipid studies of the pleural fluid should be ordered whenever the supernatant is turbid.

The persistent cloudiness of these pleural fluids after centrifugation is due to their high lipid content, which can result from one of three mechanisms. First, the lymphatic duct may be disrupted so that chyle accumulates in the pleural space. The patient is then said to have a chylothorax, and the pleural effusion is called a *chylous pleural effusion*. Second, large amounts of cholesterol or lecithin–globulin complexes can accumulate for unknown reasons in the pleural fluid. The patient is then said to have a pseudochylothorax and a chyliform pleural effusion. Although some authors have separated pseudochylothoraces into chyliform pleural effusions characterized by high lecithin–globulin levels and pseudochylous effusions characterized by the presence of cholesterol crystals (319,320) I see no advantage in making this differentiation. Third, if the patient is receiving parenteral nutrition through a central line and the superior vena cava is perforated, the fat emulsion can collect in the pleural space. It is important to distinguish these three entities because the treatment is different for each one.

When milky pleural fluid is discovered, the first question that should be asked is whether the patient is receiving parenteral nutrition through a central line. If the superior vena cava is perforated in such a setting, the intravenous infusion fluid may accumulate rapidly in the pleural space (321). If the patient is receiving a lipid emulsion, the pleural fluid triglyceride level can be very high. However, the pleural fluid glucose and the potassium levels are usually exceedingly high if the fat emulsion is entering the pleural space (321).

The diagnosis of chylothorax is best made by measuring the triglyceride levels in the pleural fluid. If the pleural fluid triglyceride level exceeds 110 mg/dL, the patient probably has a chylothorax; if the triglyceride level is below 50 mg/dL, the patient does not have a chylothorax (Fig. 7.8). If the triglyceride level is between 50 and 110 mg/dL, the patient may or may not have a chylothorax (322). When the diagnosis is uncertain, lipoprotein analysis of the fluid should be ordered. The demonstration of chylomicrons in the pleural fluid with lipoprotein analysis is diagnostic of chylothorax (322,323). Most patients with chylothoraces and pleural fluid triglyceride levels below 110 mg/dL are malnourished. Except for diagnosing chylothorax, pleural fluid triglyceride levels have no role in the diagnosis of pleural effusions (324).

Romero et al. (325) have suggested that the diagnosis of chylothorax should also require a pleural fluid-to-serum cholesterol ratio lower than 1 and a pleural fluid-to-serum triglyceride ratio less than 1. The first requirement is reasonable because patients

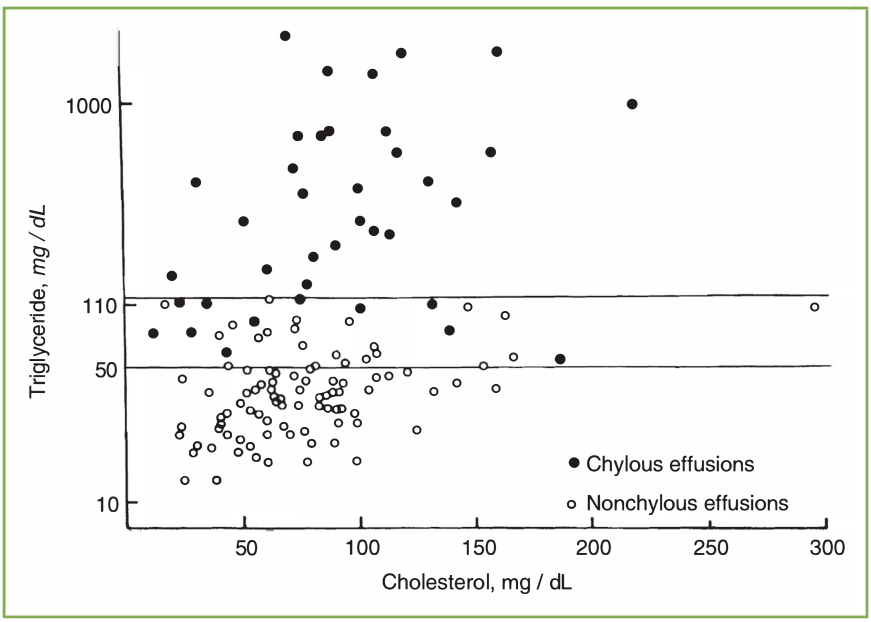

FIGURE 7.8 ■ Scattergram of cholesterol and triglyceride values in pleural fluid. Solid dots represent patients whose effusions contained chylomicrons and were considered chylous. Open circles represent patients without chylomicrons whose effusions were considered nonchylous. All patients with triglyceride levels above 110 mg/dL had chylous effusions, whereas no patient with triglyceride levels below 50 mg/dL had a chylous effusion. *(From Staats BA, Ellefson RD, Budahn LL, et al. The lipoprotein profile of chylous and nonchylous pleural effusions.* Mayo Clin Proc. *1980;55:700–704, with permission.)*

with chyliform pleural effusions often have pleural fluid cholesterol levels above 250 mg/dL (326) and this requirement should eliminate them. It should be noted that the clinical picture with chyliform effusion of a long-standing pleural effusion with thickened pleura should be easy to differentiate from that of an acute pleural effusion with normal pleural surfaces seen with chylous effusion. If any doubt exists, lipoprotein analysis of the pleural fluid should be performed. The second criteria proposed by Romero et al. (325) is probably not necessary because in patients without chylothorax, there is no relationship between the serum and the pleural fluid triglyceride level, and the mean pleural fluid triglyceride level is only approximately 25% of the serum level (32).

Other studies useful for diagnosing turbid pleural fluids are the total lipid content, the cholesterol content, and microscopic examination of the sediment. Most pleural effusions that are cloudy secondary to high lipid levels have a total fat content greater than 400 mg/dL (327). The cholesterol levels in the pleural fluid are elevated in high-lipid pleural effusions because of high numbers of cholesterol crystals or lecithin–globulin complexes (320), but cholesterol levels may also be elevated in chylous pleural effusions (322), as demonstrated in Figure 7.8. When the turbidity is due to high numbers of cholesterol crystals, examination of the pleural fluid sediment reveals the cholesterol crystals, which are large, rhombic, or polyhedric, as illustrated in Figure 26.1.

MICROBIOLOGIC STUDIES ON PLEURAL FLUID

Cultures

Pleural fluid from patients with undiagnosed exudative pleural effusions should be cultured for bacteria (both aerobically and anaerobically), mycobacteria, and fungi. For aerobic and anaerobic bacterial cultures, the pleural fluid should be inoculated directly into blood culture media at the bedside because the number of positive cultures will increase with this method (328,329). In one study (329) of 62 patients with suspected pleural infection, the addition of the blood culture bottle culture to the standard culture increased the proportion of patients with identifiable pathogens from 37.7% to 58.5%. If the fluid is not inoculated at the bedside, it should be sent to the laboratory in an anaerobic transport container at room temperature (330). A Gram's stain of the fluid should also be obtained.

For mycobacterial cultures, use of a BACTEC system with bedside inoculation provides higher yields and faster results than do conventional methods. In one study, the median time for the BACTEC cultures to become positive was 18 days (range 3–40 days), whereas the median time for conventional cultures was 33.5 days (range 21–48 days) (331). Routine smears for mycobacteria are not indicated because they are almost always negative, unless the patient has a tuberculous empyema or unless the patient is HIV positive.

It has been suggested that microbiologic testing of pleural fluid should be ordered more selectively (332). The basis for this recommendation was the observation at the Mayo Clinic that only 15 of 476 patients (3%) had positive cultures (332). The results of the study by Ferrer et al. (333) tend to go against this conclusion. They reviewed the results of 245 pleural fluid specimens inoculated into blood culture vials and found that 15% of the specimens were positive and 60% of the positive specimens occurred in fluid that was nonpurulent (333). Certainly, it is not cost effective to obtain cultures from transudative pleural fluids, but I believe cultures should be obtained from most patients with exudative effusions of unknown etiology.

Culturing pleural fluid from chest tube drainage should be discouraged. Cultures from chest tubes yield inaccurate culture results when compared with direct aspirates (330).

Nucleic Acid Amplification (NAA)

It has been shown that using NAA to amplify and sequence the bacterial ribosomal RNA gene, that the bacteria responsible for a complicated parapneumonic effusion can frequently be identified (334). Maskell et al. (334) performed this procedure on 404 pleural fluid specimens obtained during the First Multicenter Intrapleural Sepsis Trial (335). They reported that the NAA technique identified bacteria in 70 samples which were negative on culture (335). The standard cultures and the nuclear amplification technique were complimentary in identifying different organisms when they were both positive (334).

Countercurrent Immunoelectrophoresis

The goal of countercurrent immunoelectrophoresis (CIE) is to identify bacterial antigens in pleural fluid, and to thereby establish a presumptive bacteriologic diagnosis in patients with parapneumonic pleural effusions. CIE depends on the interaction of an antigen with a negative charge and a specific antibody with a positive charge in an electrical field to form a distinct precipitin line (336,337). The advantage of CIE over bacterial cultures is that results are available within hours rather than days, so that appropriate antibiotics can be administered sooner. CIE is most useful in the diagnosis of pleural effusions in children, in whom most pleural effusions are due to bacterial infections with *Streptococcus pneumoniae, S. aureus,* or *Hemophilus influenza.* These are the three bacteria for which antigens are available to perform CIE studies. In a series of 87 pediatric patients with pleural effusions (336), pleural fluid cultures were positive for one of these three bacteria in 34 patients, and CIE correctly identified the offending organisms in 33 (97%). In approximately 25% of those patients, the Gram's stain of the pleural fluid was negative. In an additional 23 patients, CIE identified antigens in the pleural fluid when the pleural fluid cultures were negative (336).

Another advantage of CIE is that it remains positive for several days after antibiotic therapy is initiated (332,333). To my knowledge, examination of pleural fluid with CIE has not been performed on a large series of adult patients with pleural effusion. The disadvantage of CIE in the adult patient with a complicated parapneumonic effusion is that many such effusions are due to anaerobic bacteria (117), and antigens for all the anaerobic organisms are not yet available for routine use. Certainly, if CIE is readily available, it should be performed on the pleural fluid from patients with an acute febrile illness and pleural effusion.

Direct Gas–Liquid Chromatography

Most anaerobic bacteria produce volatile fatty acids. Demonstration of such fatty acids by direct gas–liquid chromatography of the pleural fluid has been proposed as a means to establish the diagnosis of anaerobic pleural infections. In one report, pleural fluid from 52 patients, including 14 with anaerobic infections, were analyzed with direct gas–liquid chromatography (338). Multiple volatile fatty acids or succinic acid was present in 13 of the 14 (93%) fluids from patients with anaerobic infections. Succinic acid was the major product in 10 patients, and 9 of them had infections due to *Bacteroides.* In contrast, pleural fluid from other patients did not contain multiple volatile fatty acids or succinic acid (338). Direct gas–liquid chromatography is a difficult and expensive procedure, however, and it requires specially trained technicians. The resources required for gas–liquid chromatography are probably better spent in upgrading anaerobic culture techniques.

Fibrinogen and Fibrin Degradation Products

The fibrinogen levels in pleural fluid are low in comparison to those in plasma (339,340). The levels of fibrinogen tend to be low in patients with CHF and malignancy, and high in patients with tuberculosis and empyema but there is much overlap. The level of D-dimer is comparable in all pleural effusions (340).

MISCELLANEOUS TESTS ON PLEURAL FLUID

Other Proteins

Numerous studies have evaluated the diagnostic usefulness of measuring other proteins in the pleural fluid, including, ferritin (341), mucoproteins (112), fibronectin (342), acid glycosaminoglycans (mucopolysaccharides) (343), β_2-microglobulins (344), and α-fetoprotein (344). The pleural fluid levels of these proteins are not useful in the differential diagnosis of exudative pleural effusions.

Other Enzyme Determinations

Many other enzymes, including aldolase (345), glutamic oxaloacetic transaminase, glutamic pyruvic transaminase (345), phosphohexose isomerase (345), malic dehydrogenase (345), isocitric dehydrogenase (345), glutathione reductase (345), alkaline phosphatase (346), angiotensin-converting enzyme (347), and transketolase, have been measured in the pleural fluid and have been found to give no useful diagnostic information. One report suggested that elevation of the acid phosphatase level in the pleural fluid was diagnostic of metastatic prostatic carcinoma (348), but a second report indicated that approximately 10% of all pleural effusions have acid phosphatase levels above the upper normal limit for serum and higher than those in the corresponding serum (349).

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